

Which state are you converging? A spin audit of sample-based quantum diagonalization benchmarks on iron–sulfur clusters

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Draft v6.0 — September 11, 2026

Abstract

Sample-based quantum diagonalization (SQD/QSCI) diagonalizes the Hamiltonian classically in determinant subspaces selected from quantum-circuit samples. Its iron–sulfur demonstrations underpin claims of utility-scale quantum chemistry, and a recent critique disputes them at matched subspace dimension. Both sides compare energies; neither reports which spin state was computed. Our spin audit uses the exact determinant-space $\langle S^2 \rangle$, the invariant spectrum of the S^2 block over quasi-degenerate manifolds, and higher moments ($\langle S^4 \rangle$ to $\langle S^8 \rangle$) bounding the spin-sector weights. We apply it to the public [2Fe-2S] and [4Fe-4S] benchmarks, IBM’s released `qiskit-addon-sqd` pipeline and published hardware data. No audited execution returned the named singlet at competitive energy; for published states unavailable to direct audit, the energy-only records do not establish singlet identity. Under the audited HCI protocols, selected-CI growth approaches a spin-contaminated state ($\langle S^2 \rangle \approx 4.7$ on [2Fe-2S]) whose samples keep the contamination. Spin-inversion completion leaves the energy and $\langle S^2 \rangle$ of these states unchanged. The released [2Fe-2S] hardware samples give no energy advantage over the released uniform-random control. A scalar mean does not identify these states: two audited arms share $\langle S^2 \rangle \approx 2.00$, but one is a triplet eigenstate ($\text{Var}(S^2) = 3 \times 10^{-6}$) and the other a 2:1 singlet–quintet mixture whose third moment bounds its triplet weight below 10^{-4} . At the largest published dimension (5.625×10^7 determinants, reconstructed here from the released support), the lowest root, 170 mHa below the published energy, has $\langle S^2 \rangle = 1.3711$, unchanged under an 18.6-fold residual tightening to 1.1% of the 0.285 mHa gap. Its first two moments, with $\text{Var}(S^2) = 6.9$, exclude any decomposition confined to $S \leq 2$. Near-singlet character appears only as a non-stationary superposition, because determinant selection does not commute with S^2 . Energy-only benchmarks therefore do not establish convergence to the named state. We recommend reporting spin moments with every energy and controlling spin during selection.

Principal findings

- Energy-only SQD and QSCI benchmarks do not establish which spin state was computed. On the iron–sulfur benchmark systems, misidentifying the spin state shifts the energy on the same scale as the effects under dispute.
- IBM’s released [2Fe-2S] hardware samples give no energy advantage over the released uniform-random control in any audited comparison.
- Spin-inversion completion does not control S^2 . On the benchmark-protocol states it leaves the ground state unchanged, and the one spin-pure state it produced (from the [4Fe-4S] hardware samples) is a triplet eigenstate to measured tolerance ($\text{Var}(S^2) = 3 \times 10^{-6}$).

- At the largest published [2Fe-2S] dimension (5.625×10^7 determinants, reconstructed here from the released support), the certified lowest root has $\langle S^2 \rangle = 1.3711$, unchanged under an 18.6-fold residual tightening to 1.1% of the ground gap. Its variance is $\text{Var}(S^2) = 6.9$, and its spin moments require support beyond $S = 2$.
- Within the three lowest roots of that subspace, near-singlet character exists only as a non-stationary superposition. Determinant selection does not commute with S^2 ($[PHP, PS^2P] \neq 0$), so the eigenbasis of the projected Hamiltonian need not align with spin sectors.
- A scalar $\langle S^2 \rangle$ does not identify a state. Two audited arms both have $\langle S^2 \rangle \approx 2.00$. One is a triplet eigenstate (LP triplet weight = 1.000 over all sectors; $\text{Var}(S^2) = 3 \times 10^{-6}$). The other has its triplet weight bounded below 10^{-4} across all sectors by its third moment ($\text{Var}(S^2) = 8.0$; a 2:1 singlet–quintet mixture, $w_0 = 0.665 / w_2 = 0.335$).
- A spin penalty applied at diagonalization does not rescue the audited benchmark subspaces; the deficit is upstream in the subspace. We applied the penalty ($ss = 0$) to the subspace that reaches benchmark energy in the package’s linear form, at and above its default strength, and in the flagship’s stated squared form at the stated $\lambda = 0.2$. We ran the squared form in either compression onto the subspace. The projected square holds $\langle S^2 \rangle = 3.01$ at +528 mHa, and the literal square $P(S^2)^2P$ reaches a singlet only at the +2.9 Ha floor. In every form the singlet end of the penalty frontier lies 2.9 Ha above the reference, so the sampled subspace contains no competitive singlet, and the package’s own penalty path does not reach the frontier from a cold start. The fix belongs in spin-aware selection (CSF bases, or S^2 -projected selection that keeps the recoupling determinants), with measured spin moments reported alongside every energy.

1 Introduction

Two lines of work define the current sample-based quantum diagonalization (SQD, also called QSCI) literature, and the disagreement between them has so far been argued from energies alone.

The first is the utility-scale program exemplified by IBM’s iron–sulfur demonstrations (arXiv:2405.05068; Sci. Adv. 2025). Samples from a local unitary cluster Jastrow (LUCJ) circuit run on quantum hardware select determinant subspaces of dimension up to $\sim 10^8$ for [4Fe-4S]-class active spaces, and the Hamiltonian is then diagonalized in those subspaces on classical HPC. The resulting energies are presented as competitive electronic-structure results for systems “beyond the scale of exact diagonalization.” The program has continued to scale: a RIKEN–IBM successor (arXiv:2511.00224, November 2025) reports closed-loop SQD at subspace dimensions up to 10^{10} on the Fugaku supercomputer, with improved energies for the same [4Fe-4S] cluster. We discuss it in Section 3.

The second is the critique of Reinholdt and co-workers (arXiv:2501.07231, “Critical Limitations in Quantum-Selected Configuration Interaction Methods”). At matched subspace dimension, which is the first-order cost-fair comparison because diagonalization dominates the classical cost, they found that determinants selected from quantum samples never outperform determinants selected by classical heuristics such as heat-bath CI (HCI) or CIPSI. On $N_2/\text{cc-pVDZ}$, QSCI needed orders of magnitude more samples to approach the accuracy of HCI and did not surpass it at matched subspace size.

We compared the two positions under one protocol. Every arm of every comparison runs through the same recovery and diagonalization code, every reported subspace size is the dimension actually diagonalized, and every Hamiltonian is checked by three-way agreement (our Slater–Condon engine, PySCF and PyCI agree to μHa on test systems). The results fall into three parts.

First, the critique replicates, with its authors’ tooling and with ours, on their system and on new ones. Among equilibrium geometries, stretched bonds and a diradical transition state, we found no regime in which circuit samples beat classical selection at matched determinant count (Section 2.1).

Second, an earlier round of this work produced an apparent counterexample, which we have retracted. At a diazene rotation transition state, classically seeded selected CI appeared to stall ~21 mHa above the exact energy, while sampling the exact eigenvector’s distribution did not. A spin audit showed that the stalled root was a triplet, so the comparison had crossed spin sectors. Compared within one spin sector, classical selection is again more accurate (+0.71 vs +1.93 mHa at $D = 4096$). We report the retraction in full (Section 2.2) because its lesson is general: in near-degenerate spectra with several spin states at low energy, an energy-only benchmark cannot show which state was converged, and the outcome of a comparison can depend on exactly that.

Third, the iron–sulfur clusters behind the utility-scale claims are especially exposed to this problem. Their low-energy spectrum is a Heisenberg ladder of spin states spanning ~15–30 mHa, the same order as the accuracy improvements being claimed, and the $M_S = 0$ determinant sector contains every spin state of the ladder, even and odd. We are not aware of any published SQD or QSCI study, on these systems or any other, that reports the measured determinant-space $\langle S^2 \rangle$ of the states it obtained. This includes the November 2025 RIKEN–IBM successor (arXiv:2511.00224), which reports energies and energy variances at subspace dimensions up to 10^{10} ; the words “singlet” and “triplet” each occur once in its full text, inside the title of a cited reference. It also includes the variants with convergence guarantees (randomized sample-based Krylov diagonalization, arXiv:2508.02578), which likewise report energies without state identity. The critique frames convergence in terms of good and bad initial guesses, also without spin resolution.

The flagship work does name its target, the singlet, and states two spin mechanisms: determinant completion, and “conservation of total spin by a soft constraint in the eigenstate solver, i.e. by adding a penalty term to mitigate spin contamination” ($H + \lambda[S^2 - s(s + 1)]^2$, employed at $\lambda = 0.2$). It reports the $\langle S^2 \rangle$ achieved by neither. Spin penalties are known in this literature. Besides the flagship’s stated constraint, a quantum-centric methylene study (arXiv:2411.04827) applies a `fix_spin`-style Lagrange term (targeting CH_2 states via particle-number sectors), and a 2026 cluster-adaptive SQD preprint (arXiv:2603.09346) penalizes S^2 on the same [2Fe-2S] active space. None of the three reports the $\langle S^2 \rangle$ achieved, and the released pipeline leaves the penalty off by default. We measure what is not reported (Sections 2.3–2.6), including the stated penalty, in the package’s linear form and in the flagship’s stated squared form at its stated strength (Section 2.6). The states returned by the audited executions are not singlets; with a single measured exception they are not spin eigenstates at all, and that exception is a triplet. On these systems, both sides of the dispute were comparing energies of unidentified states.

Our main conclusion concerns benchmark practice rather than either side of the dispute. At the accuracy scales in question, the identity of the computed state determines what an energy means, and it is not currently measured. Measuring it is cheap: a determinant-space $\langle S^2 \rangle$ costs one sparse matrix–vector product, and spin-penalty solvers already exist in the released toolboxes (Section 2.6). Until it is routine, a claim of chemical accuracy on [4Fe-4S] is a claim about an energy, and not yet about a named electronic state.

2 Results

2.1 Sampled versus selected subspaces at matched dimension

We reproduced the critique’s central N_2 (10e, 22o) comparison with the authors’ PyCI-based tooling. At matched determinant count, QSCI-style selection from state samples approaches HCI from above, with only a nominal -0.6 mHa crossing at the largest budget. As the shot budget grows from 10^5 to 10^7 the energy gap goes $+3.1 \rightarrow +0.5 \rightarrow -0.6$ mHa, and each tenfold increase in samples adds only ~3.5 times as many retained determinants. At stretched geometry ($R = 2.0$ Å) the pattern recurs, with a nominal edge of -1.1 mHa at the

largest budget. That edge is a proxy artifact. The samples were drawn from HCI’s own converged vector, and sampling a vector can rediscover the selection that built it but cannot improve on it.

On diazene (N_2H_2) stationary points, including the rotation transition state, a diradical where single-reference methods fail, the spin-consistent comparison (Section 2.2) gives +0.71 mHa for CIPSI and +1.93 mHa for exact-distribution sampling at $D = 4096$, over three RNG seeds. The sampling arm here is idealized: its samples are drawn from the exact eigenvector’s $|c|^2$ distribution, which is the best any circuit reproducing this state could do. Realistic circuit sampling is much worse, and the reason lies in the distribution itself. Two million shots from the exact diradical eigenvector yield only 2,869 unique determinants, because most of the weight sits on a few leading configurations. A CCSD-initialized LUCJ circuit, the flagship’s initialization, yields 181 unique determinants from 2×10^6 shots and lands at +64 mHa. Optimizing the LUCJ parameters variationally improves this only to +35 mHa, still ~ 50 times the CIPSI error at the same subspace size. Sampling noise adds spurious diversity and erratic energies. The saturation of unique determinants is a property of concentrated eigenvector distributions and would occur on noiseless hardware.

The unique-determinant yield of any fixed sampling distribution has the closed form $E[U(M)] = \sum_i [1 - (1 - p_i)^M]$. This is an exact occupancy identity for independent computational-basis shots of a fixed prepared state, which is the operation QSCI defines. Adaptive preparations, measurement-basis changes and amplitude-estimation access lie outside it, and outside every claim built on it here. We evaluated the formula on the measured distributions of this study and compared it with sampled unique counts wherever a measurement exists. For the converged [2Fe-2S] benchmark vector it gives 38,129 uniques at 10^6 shots vs measured 38,118 (0.03%). For the N_2 equilibrium and stretched HCI vectors the maximum deviations are 1.4% and 0.2% across 10^5 – 10^7 shots, and for the noiseless [2Fe-2S] LUCJ distribution built from IBM’s circuit template it gives 5,344 vs 5,338 sampled (0.1%). The local exchange rate $\gamma = d \log U / d \log M$, the number of new configurations another decade of shots buys, is 0.39–0.57 at every operating point measured, so ten times the shots yields 2.4–3.7 times the configurations. The yield also saturates. For the 170,448-determinant [2Fe-2S] benchmark state, even 10^8 shots recover an expected 131,667 of its determinants, so sampling does not reproduce the state’s own support at any realistic budget (Fig. 1).

At matched diagonalized dimension, then, we find no regime (equilibrium, strongly correlated or diradical) in which quantum-circuit samples beat classical selected CI. This confirms and extends arXiv:2501.07231. The comparison also sets a ceiling. For a given prepared state, the exact-distribution arm is the noiseless, device-independent sampling limit: no quantum computer sampling a fixed prepared state in the computational basis (the operation QSCI defines) can do better than its own $|c|^2$ distribution, and at matched D that limit loses. The bound applies to samplings of a fixed state and says nothing about the choice of state. A deliberately different preparation, with flatter amplitudes or up-weighted recoupling determinants, is a separate lever that we did not test and make no claim about; so are adaptive preparations, measurement-basis changes and amplitude-estimation schemes, which lie outside this protocol.

The accuracy ordering does not depend on pricing the samples. Matched D is a cost model, and the conclusion does not rest on it. The dominant cost of SQD is the classical eigensolver call, as IBM’s package documentation states, and solve cost is monotone in D . At every matched D measured here, sampled subspaces are less accurate than selected ones (the gaps of +1.1 to +1.2 mHa above), at any shot price including zero. Where sampling does converge toward the classical curve (N_2), the determinant count needed for matched accuracy is parity at best. The measured inflation is 0.86–1.38 $\times$, and values below parity arise only when the sampler draws from the classical solver’s own converged vector, the proxy artifact described above. The cost of the shots comes on top. No relative pricing of shots against CPU time changes the accuracy ordering at matched D . Total wall-clock time at fixed D also depends on subspace structure and solver implementation (the successor work’s half-string tensor-product spaces, for instance, admit specialized Hamiltonian kernels); our claim concerns the accuracy ordering at matched dimension, not total run time.

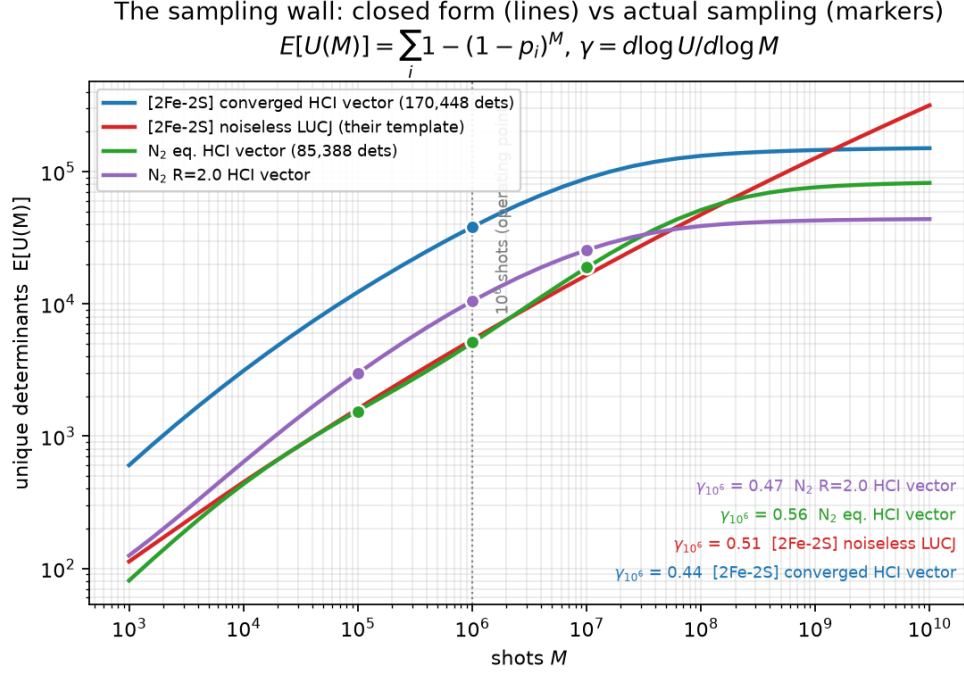


Figure 1: Unique-determinant yield against shot count: the closed form $E[U(M)] = \sum_i 1 - (1 - p_i)^M$ (lines) and sampled unique counts (markers) for the measured distributions of this study, with $\gamma = d \log U / d \log M$ marked at the 10^6 -shot operating point.

2.2 A retracted result at the diazene transition state

In the first round of this work, results at the diazene rotation transition state seemed to contradict the conclusion above. Every classically seeded arm (CIPSI and HCI from Hartree–Fock; HCI from CISD) stalled at +21–23 mHa even at $D = 4096$, which is 9% of the full sector, while exact-distribution sampling descended smoothly to +1.9 mHa at the same D . We first attributed this to selection hysteresis: selected CI grows around its current eigenvector and cannot move to a different one at a near-degeneracy.

The spin audit confirmed the hysteresis and showed that the comparison was flawed. At this geometry the lowest $M_s = 0$ roots are a near-degenerate singlet–triplet pair, and the classically grown spaces were converging to the triplet while being scored against the singlet reference. Scored consistently, with each arm compared against the eigenstate it actually follows or with all arms constrained to the singlet, classical selection is more accurate at every matched D (CIPSI +0.71 vs sampling +1.93 mHa at 4096). The same audit invalidated a mechanism claim in a related project of ours, a “correlation flips the mechanism” result that rested on the contaminated root, and we retracted that too.

The episode set the method for the rest of the audit: every stationary point in this literature needs a spin characterization before its energy has a state-specific meaning. It also set the order of work. The diradical was the strongest candidate we knew of for a quantum-sampling advantage, and it did not survive spin resolution, so the same test had to be applied to the flagship systems before their claims could be assessed.

2.3 Rotation-invariant $\langle S^2 \rangle$ for selected-CI manifolds

In the $M_s = 0$ sector, $S^2 = S^- S^+ + S_z(S_z + 1)$ reduces the spin measurement to the Gram matrix $G_{ij} = \langle S^+ v_i | S^+ v_j \rangle = \langle v_i | S^2 | v_j \rangle$ over any set of eigenvectors $\{v_i\}$. The measurement has two properties that matter

here.

1. It is exact in the selected space. No RDM approximation is involved: the cost is a sparse application of S^+ followed by a Gram accumulation, one pass over the determinant pairs connected by single spin flips.
2. It is invariant under rotations within a quasi-degenerate manifold. Where eigenvalues are degenerate or nearly so, the $\langle S^2 \rangle$ of an individual eigenvector depends on which rotation within the manifold the solver happens to return. The eigenvalues of the S^2 block over the cluster do not, and they answer the question a singlet-target mitigation claim raises: does the manifold contain a singlet direction at all? For a singlet the test is sharp, because S^2 is positive semidefinite and a zero expectation value therefore forces $S^2 v = 0$. For higher spin a matching $\langle S^2 \rangle$ is not sufficient, and the $\langle S^4 \rangle$ measurement introduced below tests whether the variance vanishes. Roots are grouped into clusters when consecutive gaps fall below $2 \mu\text{Ha}$ (Methods). The choice does not affect the results: re-deriving every partition in this paper at 0.2, 2 and $20 \mu\text{Ha}$, and under a residual-adaptive criterion, changes none of the paper’s reported clusters or block spectra (Methods, threshold scan; `_thrscan.py`).

A scalar $\langle S^2 \rangle$ bounds the spin decomposition but does not determine it, since many sets of sector weights share one expectation value. We therefore use it only for incompatibility statements (a singlet has $\langle S^2 \rangle = 0$) and report Gram spectra where manifold resolution matters. Spin is also only the first test of state identity. A spin-pure root can still be the wrong root of its sector, so spin moments are necessary but not sufficient; the full set of checks (spatial symmetry, particle number, and overlap or root continuity against a declared target) is listed in the Recommendations.

We validated the measurement against PySCF spin labels on diazene (Gram diagonal 2.000000, 0.000000, 0.000000 for a triplet/singlet/singlet triple; off-diagonals $< 4 \times 10^{-12}$), and every production run is gated on this check. The gate caught a uint64 sign-wrap defect in an earlier revision of our code before any affected number was recorded.

2.4 [2Fe-2S]: selected-CI growth from two seeds

We audited the public Li–Chan [2Fe-2S] active space (30e, 20o; $M_s = 0$ sector dimension 2.4×10^8 ; near-exact singlet reference $E = -116.6056091 \text{ Ha}$). Determinant spaces were grown by HCI ladders from the two kinds of seed the controversy discusses: a broken-symmetry (BS) two-determinant seed, the “good guess”, and a closed-shell aufbau seed, the “bad guess”. Table 1 lists the energy and $\langle S^2 \rangle$ of the followed sector root (see also Fig. 2 and Fig. 3).

Table 1: [2Fe-2S] followed sector-root energies and spin content vs subspace size D .

Seed	D	$E_0 \text{ (Ha)}$	err (mHa)	$\langle S^2 \rangle \text{ root0}$	$\langle S^2 \rangle \text{ root1}$
BS	1,011	-116.456202	+149.4	4.978	4.977
BS	25,734	-116.569067	+36.5	4.893	4.921
BS	65,124	-116.584165	+21.4	4.779	4.863
BS	155,030	-116.592307	+13.3	4.659	4.809
aufbau	1,003	-116.060455	+545.2	3.033	3.015
aufbau	5,053	-116.155646	+450.0	2.050	2.073
aufbau	24,570	-116.430693	+174.9	4.672	4.461
aufbau	68,825	-116.579203	+26.4	4.824	4.876
aufbau	170,448	-116.591986	+13.6	4.669	4.810

The followed state is never a spin eigenstate. Its $\langle S^2 \rangle$ of 4.7–5.0 lies between the $S = 1$ and $S = 2$ values (2 and 6), so it is a mixture of spin sectors: neither a slightly contaminated singlet nor a clean high-spin state.

The BS seed starts at exactly $S^2 = 5.000$, as constructed, and decays very slowly. The aufbau seed wanders ($3.0 \rightarrow 2.05 \rightarrow 4.7$) before rising into the same range. Root 1 is equally mixed (4.8–4.9 at every checkpoint), so neither of the two lowest roots is close to a spin eigenstate.

The two seeds converge to the same state. By $D \approx 1.6 \times 10^5$, under the HCI protocols audited here, they agree to 0.3 mHa in energy and 0.01 in $\langle S^2 \rangle$. The framing in which convergence quality depends on the initial guess therefore does not apply at this scale: the selection dynamics determine what is converged, independently of the seed, and what they converge is the spin mixture favored by compact determinant expansions. Once each arm has reached this state, the decrease of $\langle S^2 \rangle$ per doubling of D is roughly constant (0.085–0.097 per doubling for the BS arm over its final three intervals; 0.106–0.133 for the deep aufbau arm), with no sign of acceleration. At that rate the trend cannot produce a singlet within the sector. Even at the fastest measured rate, reaching $\langle S^2 \rangle < 1$ would take ~ 28 further doublings, to a dimension ($\approx 3 \times 10^{13}$) five orders of magnitude larger than the entire 2.4×10^8 sector. The exact singlet is of course recovered at the full sector. What the data show is that no approach to it is visible anywhere in the observed range, so the recovery must happen in a final regime about which the measured trend says nothing (1.55×10^5 determinants is already 0.065% of the sector).

The remaining error is the same size as the physics of interest. Both seeds stall +13 mHa above the exact singlet, and the whole Heisenberg exchange ladder of [2Fe-2S] (the singlet–triplet– $\dots$ – $S = 5$ splittings that define its magnetochemistry) spans ~ 15 –30 mHa. A state of unidentified spin with an error the size of the ladder cannot resolve the ladder. At these system sizes the chemistry of interest and the spin-identity error lie on the same energy scale, which is why state identity has to be established before an energy can be interpreted.

The mechanism is the known compactness bias of truncated CI toward high spin. High-spin components can be represented with fewer determinants, so incremental selection over-rewards them. The size of the effect, and its persistence to 1.7×10^5 determinants on the flagship benchmark systems, had not previously been measured.

2.5 [4Fe-4S]: selected-CI growth

The same audit on the public [4Fe-4S] active space (54e, 36o; sector dimension $\sim 8.8 \times 10^{15}$; approximate DMRG reference -327.2396369 Ha) gives what is, to our knowledge, the first spin resolution of selected-CI states on this system (Table 2).

Table 2: [4Fe-4S] followed sector-root energies and spin content vs subspace size D .

D	E_0 (Ha)	err (mHa)	$\langle S^2 \rangle$ root0	$\langle S^2 \rangle$ root1
1,106	−326.184317	+1055.3	5.293	5.270
5,905	−326.385018	+854.6	6.053	5.914
17,993	−326.445889	+793.7	6.464	6.302
40,435	−326.484822	+754.8	7.004	9.284

Here $\langle S^2 \rangle$ increases with subspace size. Starting from a closed-shell seed ($S^2 = 0$), growth moves monotonically away from the singlet and reaches $S^2 = 7.0$, beyond the pure $S = 2$ value, at 4×10^4 determinants. The low-lying spectrum is a dense manifold of mixed spin: at $D = 40k$, root 1 has $S^2 = 9.28$ and root 2 has $S^2 = 5.84$, and adjacent roots differ by $\Delta S^2 \approx 3.4$. The energies are far from the reference (4×10^4 determinants is only 5×10^{-12} of the sector), so we rely on the trend alone. On the system where flagship SQD operates at sector fractions of $\sim 10^{-8}$, selection moves away from the named target state, and nothing measured here gives a reason to expect singlet identity there without explicit spin control.

2.6 Spin-inversion completion and spin penalties

Flagship SQD work addresses spin through configuration recovery combined with spin-inversion closure, which the literature calls spin completion: the subspace is augmented with spin-inverted determinants. In the released software this is done by symmetrizing the α and β half-string sets, which yields the full spin-inverted closure of the product space. Closure makes the subspace invariant under spin inversion and so removes one obstruction to representing spin-parity eigenvectors. It does not make the subspace S^2 -invariant, and it does not guarantee that any low-energy eigenstate is a spin eigenstate. Whether the variational ground manifold within the completed subspace contains a singlet is the question the Gram measurement answers, and we tested it in two ways.

Direct audit. At every checkpoint of every arm above (up to 319,189 completed determinants on [2Fe-2S] and 80,755 on [4Fe-4S]) we completed the space and solved it again. At every scale and on both systems, the ground energy changed by less than $1 \mu\text{Ha}$ while the determinant count roughly doubled. The ground state became an exactly degenerate pair $\{v, \bar{v}\}$, the state and its spin image, and the 2×2 S^2 block over that pair had eigenvalues $[x, x]$, where x is the original mixture value. The block is therefore proportional to the identity, and this conclusion does not depend on the basis chosen within the pair: every rotation within the completed ground manifold has the same mixed $\langle S^2 \rangle$, so the manifold contains no singlet. The added determinants receive no weight in the variational ground state. A self-contained reproducer that runs in about 35 s, `spin_inversion_is_not_singlet.py`, is in the supplementary archive. On IBM’s public MO FCIDUMP the top- N marginal-ranked α and β half-string sets are identical, which we verified at every N used in this paper including the flagship’s 7,500, so the product construction satisfies spin-inversion closure by construction. Its ground state is nevertheless strongly contaminated ($\langle S^2 \rangle = 3.50$ at $N = 60$), and the package’s own `solve_fermion/spin_square` agrees with our Gram measurement on that state to 1×10^{-4} and $0.4 \mu\text{Ha}$.

The completion mechanism at each checkpoint. Exact pair degeneracy does not follow automatically from spin-inversion symmetry, which would also allow the ground state to hybridize into energy-split symmetric and antisymmetric combinations across the original support and its image. Table 3 re-solves all 14 completion checkpoints of the three arms. It reports the quantities that decide which happens: the ground-energy shift $|\Delta E_0|$ under completion, the splitting of the completed ground pair, the invariant spectrum $\{p_1, p_2\}$ of the original-support projector compressed over that pair, the Hamiltonian coupling $\|P_+ H v\|$ from the uncompleted ground v to the added determinants, and the invariant S^2 block. A value $p_1 = 1$ with small p_2 means the pair is spanned by one state on the original support and one almost entirely on its spin image. The second eigenvalue p_2 is exactly 0 for [4Fe-4S], where the two supports are disjoint, and at most 0.11 for [2Fe-2S], where they share spin-self-inverse determinants on which the image member keeps a little projector weight. The pattern is the same at every checkpoint. The added spin-image support is uncoupled from the converged ground state at the solver’s own residual scale ($\|P_+ H v\| \leq 2.5 \times 10^{-9}$ Ha), so the ground state is unchanged: $|\Delta E_0| \leq 8 \times 10^{-13}$ Ha, with the localized pair member carrying $< 10^{-14}$ of its weight on the added determinants. Its spin image joins it as the second member of an exactly degenerate pair (splitting $\leq 1.9 \times 10^{-12}$ Ha against a solver tolerance of 10^{-9}) instead of hybridizing into split combinations. Completion thus adds the support a spin-symmetric state would need, but the variational state it was meant to repair is dynamically disconnected from that support.

One checkpoint required adjudication ($\dagger$). At the smallest [4Fe-4S] space ($1,106 \rightarrow 2,097$ determinants) the iterative Davidson solver used throughout this audit returned only one member of each degenerate pair. The completed ground state therefore looked unique, with a spurious 93 mHa “gap” to a state that is in fact the lower member of the next pair. Dense diagonalization of the full 2,097-dimensional block restores the exact structure (ground-pair splitting 0 to machine precision, $\{p_1, p_2\} = \{1.000000, 0.000000\}$, S^2 block $[5.2930, 5.2930]$), which is the same mechanism in its simplest form, and reproduces the asymmetric-space energy and $\langle S^2 \rangle$ records to all stored digits. We report the adjudication instead of silently replacing the

Table 3: Completion-mechanism audit at every spin-completion checkpoint (Section 2.6): energy shift under completion, completed-pair splitting, invariant original-support projector spectrum $\{p_1, p_2\}$ over the ground pair, coupling from the uncompleted ground to the added determinants, and the invariant S^2 block of the pair. [†]Dense-adjudicated row: the iterative solver returned one member per degenerate pair on this space (splitting reported by the dense solve). Data: `degen.npz`, `degen_dense.npz`.

space	$D \rightarrow D'$	$ \Delta E_0 $ (Ha)	pair split (Ha)	p_1, p_2	$\ P_+ H v\ $ (Ha)	S^2 block
[2Fe-2S] BS	1,011 $\rightarrow$ 2,020	2.8×10^{-13}	2.8×10^{-13}	1.0000, 0.0168	5.7×10^{-10}	[4.9783, 4.9783]
[2Fe-2S] BS	5,574 $\rightarrow$ 11,146	5.7×10^{-14}	2.6×10^{-13}	1.0000, 0.0645	3.2×10^{-10}	[4.9778, 4.9778]
[2Fe-2S] BS	25,734 $\rightarrow$ 51,466	7.1×10^{-14}	1.3×10^{-13}	1.0000, 0.0510	4.5×10^{-11}	[4.8934, 4.8934]
[2Fe-2S] BS	65,124 $\rightarrow$ 130,246	4.3×10^{-14}	2.7×10^{-13}	1.0000, 0.0441	6.1×10^{-11}	[4.7792, 4.7792]
[2Fe-2S] BS	116,270 $\rightarrow$ 232,538	2.4×10^{-13}	5.1×10^{-13}	1.0000, 0.0411	9.9×10^{-12}	[4.6978, 4.6978]
[2Fe-2S] BS	155,030 $\rightarrow$ 310,058	4.3×10^{-13}	4.3×10^{-13}	1.0000, 0.0398	2.0×10^{-11}	[4.6592, 4.6592]
[2Fe-2S] aufbau	33,632 $\rightarrow$ 45,557	2.0×10^{-13}	3.8×10^{-13}	1.0000, 0.1085	9.7×10^{-10}	[4.9545, 4.9545]
[2Fe-2S] aufbau	71,412 $\rightarrow$ 121,117	3.0×10^{-13}	3.4×10^{-13}	1.0000, 0.0511	2.5×10^{-9}	[4.8095, 4.8095]
[2Fe-2S] aufbau	109,151 $\rightarrow$ 196,595	7.4×10^{-13}	5.0×10^{-13}	1.0000, 0.0476	6.4×10^{-10}	[4.7452, 4.7452]
[2Fe-2S] aufbau	170,448 $\rightarrow$ 319,189	9.9×10^{-14}	1.4×10^{-14}	1.0000, 0.0460	7.5×10^{-10}	[4.6688, 4.6688]
[4Fe-4S]	1,106 $\rightarrow$ 2,097 [†]	5.7×10^{-14}	0	1.0000, 0.0000	3.1×10^{-16}	[5.2930, 5.2930]
[4Fe-4S]	5,905 $\rightarrow$ 11,695	3.4×10^{-13}	1.7×10^{-13}	1.0000, 0.0000	1.7×10^{-10}	[6.0527, 6.0527]
[4Fe-4S]	17,993 $\rightarrow$ 35,871	8.0×10^{-13}	1.9×10^{-12}	1.0000, 0.0000	1.5×10^{-10}	[6.4644, 6.4644]
[4Fe-4S]	40,435 $\rightarrow$ 80,755	5.7×10^{-14}	2.3×10^{-13}	1.0000, 0.0000	2.9×10^{-10}	[7.0044, 7.0044]

row because it illustrates the general point: on degenerate manifolds, iterative eigensolver output does not certify itself, and manifold measurements must be checked against an independent solve (`degen.npz`, `degen_dense.npz`).

The released pipeline. To test whether these results depend on our reimplemention of the mitigation, we ran IBM’s `qiskit-addon-sqd` pipeline unmodified: its `diagonalize_fermionic_hamiltonian` driver, post-selection, configuration-recovery loop, Cartesian-product subspace construction, `symmetrize_spin` option and `SCISState.spin_square()` diagnostic. The input was 10^6 samples drawn from our converged [2Fe-2S] vector ($D = 170,448$, $\langle S^2 \rangle = 4.669$), which is the kind of state a benchmark following this protocol produces. Table 4 gives batch means over 3 batches (`samples_per_batch=500`, 4 recovery iterations).

Table 4: The released `qiskit-addon-sqd` pipeline on 10^6 samples of the converged [2Fe-2S] HCI vector (batch means; `symmetrize_spin=True`).

Iteration	subspace	E (Ha)	err (mHa)	their $\langle S^2 \rangle$
1	$\sim 59,000$	-116.513	+92.8	4.90
2	$\sim 114,000$	-116.542	+63.3	4.87
3	$\sim 158,000$	-116.553	+52.6	4.85
4	$\sim 193,000$	-116.560	+45.7	4.837

The best single batch reaches -116.560107 Ha (+45.5 mHa) with a package-reported $\langle S^2 \rangle$ of 4.834. Across the four configuration-recovery iterations the package’s own diagnostic reports $\langle S^2 \rangle = 4.90 \rightarrow 4.84$, while the energy improves monotonically from +92.8 to +45.7 mHa. Recovery descends within the spin-contaminated manifold without changing its spin.

Running the identical experiment with `symmetrize_spin=False` gives the same result. Every batch of every iteration agrees with the symmetrized run to $\Delta E < 10^{-9}$ Ha and $\Delta \langle S^2 \rangle \sim 10^{-11}$, although the symmetrized subspace is exactly $4.00\times$ larger (194,481 vs 48,600 determinants at iteration 4). The spin-completed determinants receive zero variational weight, which is the mechanism of the direct audit reproduced inside the released code and measured by its own diagnostic.

The pipeline’s best energy, +45.5 mHa at 1.9×10^5 determinants, is 3.3 times further from the exact singlet than the +13.6 mHa of the classical HCI space that generated the samples, at comparable size (1.7×10^5). Sampling a selected-CI state and recovering from the samples is worse than keeping the selected-CI state. The 10^6 shots contain only 38,118 unique determinants, so most of the parent space is lost, and recovery cannot rebuild it at matched dimension.

The released solver (solve_sci, backed by PySCF’s kernel_fixed_space) accepts a spin_sq argument connected to PySCF’s fix_spin_ spin penalty, which is intended to suppress non-singlet character directly. The released recovery loop leaves it at its default, None. The penalty is well known in this community. The flagship’s Methods state that “we achieve the conservation of total spin by a soft constraint in the eigenstate solver, i.e. by adding a penalty term to mitigate spin contamination,” the squared constraint $H + \lambda[S^2 - s(s+1)]^2$ employed at $\lambda = 0.2$. A methylene study with IBM co-authors (arXiv:2411.04827) uses the same Lagrange-penalty construction in an SQD context. A 2026 preprint applies it to the same [2Fe-2S] active space (arXiv:2603.09346, $\lambda = 0.1$); its best energy, -116.49986 Ha, is still +105.8 mHa above the DMRG reference distributed with the flagship’s data. A mechanism aimed at the failure we measure is therefore known, implemented, stated by the flagship as employed, and off by default in the released package, and in none of these cases is the achieved $\langle S^2 \rangle$ reported.

The spin penalty. The obvious next test is to run with the penalty on, which is also what the flagship’s Methods state was done. We took the released pipeline’s converged [2Fe-2S] subspaces, the symmetrized product spaces behind the +45.5 mHa row above, captured at recovery iterations 1 and 4 ($D = 245^2$ and 441^2). The unpenalized ground state of each space reproduces the released record and agrees with an independent PyCI diagonalization of the same determinants to $0.00 \mu\text{Ha}$. On each space we computed the ground state of $H + \lambda S^2$, the exact linear operator that PySCF’s fix_spin_(ss=0) applies at $M_s = 0$, as a function of the penalty strength λ . For every converged state we recorded the true energy $\langle H \rangle$, from the RDMs as the released solver does, and both spin moments. Table 5 is for the operating-point space, $D = 441^2$; the $D = 245^2$ space behaves identically.

Table 5: Ground state of $H + \lambda S^2$ on the released pipeline’s converged [2Fe-2S] subspace ($D = 441^2$, the +45.5 mHa row of Table 4). λS^2 is the exact linear penalty PySCF’s fix_spin_(ss=0) applies at $M_s = 0$; energies and moments are the true $\langle H \rangle$, $\langle S^2 \rangle$, and $\text{Var}(S^2)$ of each converged root.

penalty strength λ	E (Ha)	err (mHa)	$\langle S^2 \rangle$	$\text{Var}(S^2)$
0 (off, the +45.5 mHa row)	-116.560107	+45.5	4.834	1.89
0.1 (fix_spin_’s shipped default)	-116.533031	+72.6	4.165	0.83
0.2 (the flagship’s stated strength, in this linear form)	-116.494963	+110.6	3.902	0.48
0.5	-116.341427	+264.2	3.441	0.18
≥ 1 (singlet sector)	-113.711	+2895	0.0000	0.0000

The resulting trade-off has no competitive middle ground. At $\lambda = 0.1$ (the released default) and 0.2, the penalty lowers $\langle S^2 \rangle$ only from 4.83 to 3.90, while the true energy rises monotonically from +45.5 to +111 mHa. The ground state stays in the spin-contaminated range ($\langle S^2 \rangle \geq 3.4$) up to +264 mHa. The first genuine singlet, the lowest exact spin-0 eigenstate the subspace can represent ($\langle S^2 \rangle = \text{Var}(S^2) = 0.0000$, residual 10^{-8}), lies 2.9 Hartree above the reference. The $D = 245^2$ space reaches its singlet at +3.03 Ha, and the lowest closed-shell determinant in either space, itself a singlet, lies at +3.05 Ha. The monotone rise follows from the form of the penalty: λS^2 is positive semidefinite, so within a fixed subspace the penalized ground state’s true energy can only exceed the unpenalized minimum, and the singlet end of the curve is the lowest-energy exact singlet the subspace contains. No penalty strength gives a singlet at competitive energy because the sampled subspace contains none. This is consistent with marginal-product selection keeping the determinants the energy needs and omitting the spin-recoupling partners that spin identity needs. The

spin–energy trade-off of Section 2.8 is therefore a property of the sampled subspace, not of the solver or its settings, and its two ends are nearly three hartree apart (Figure 4). The same semidefiniteness settles the low-spin end without a further calculation. The low- $\langle S^2 \rangle$ hardware and template subspaces of Section 2.8 already lie +229 to +248 mHa above the reference, and a penalty cannot lower $\langle H \rangle$ below a subspace’s unpenalized minimum. None of the audited subspaces therefore yields a competitive singlet under any spin penalty: the benchmark-energy subspace by direct measurement, and the others by the semidefiniteness of λS^2 .

The flagship’s stated form, at the flagship’s stated strength, is the squared constraint $H + \lambda[S^2 - s(s+1)]^2$ with $\lambda = 0.2$; for the singlet target $s = 0$ the penalty is $\lambda(S^2)^2$. Compressing $(S^2)^2$ onto a fixed determinant subspace P is not unique. A Krylov eigensolver confined to P applies the projected square $(PS^2P)^2$, the projected operator squared inside the subspace. The literal matrix compression of $(S^2)^2$ is $P(S^2)^2P$, which lets S^2 leave P and return. The two differ by the positive-semidefinite leakage operator $P(S^2)^2P - (PS^2P)^2 = PS^2(I - P)S^2P \geq 0$, which is nonzero here for the same reason that $[PHP, PS^2P] \neq 0$: the subspace is not S^2 -invariant. The difference cannot be assumed negligible, so we built both operators and ran each on the benchmark-energy subspace at the stated $\lambda = 0.2$. The literal square uses the exact $M_s = 0$ spin ladder (the S^+ machinery of the second-moment kernel), checked operator by operator against PySCF’s independent S^2 to machine precision (`_sqdpen_true.py`, `sqdpen_true_it4.npz`; `_sqdpen_sq.py`, `sqdpen_sq_it4.npz`; each $\lambda = 0$ anchor reproduces the released record under the same gates). The two compressions bracket the outcome, and neither gives a competitive singlet.

The projected square puts the ground state at +528 mHa with $\langle S^2 \rangle = 3.010$, an 11.6-fold increase over the unpenalized error and still three units of $\langle S^2 \rangle$ from the singlet. It also shows the mechanism most clearly. The converged state is nearly an eigenvector of the projected operator PS^2P , with in-subspace variance 0.009 concentrated at 3.01, which is not $S(S+1)$ for any S . Its exact first moment alone forces a physical spin variance of at least $(\langle S^2 \rangle - 2)(6 - \langle S^2 \rangle) \approx 3.0$, because the physical S^2 spectrum is the lattice $\{0, 2, 6, 12, \dots\}$ and the minimum-variance distribution at fixed mean lies on the two adjacent points. The state is therefore sharp in PS^2P at an unphysical value and broad in S^2 itself.

The literal square $P(S^2)^2P$, which keeps the leakage the projected form drops, is the stronger penalty. It registers the unpenalized state’s full $\langle S^4 \rangle \approx 47$ instead of its in-subspace value of ≈ 23 , and at the same $\lambda = 0.2$ it drives the ground state past a sharp threshold to a singlet at the subspace floor: $\langle S^2 \rangle = 0.004$, $\text{Var}(S^2) = 0.03$, at +2,884 mHa. The $D = 245^2$ space reaches $\langle S^2 \rangle = 0.000$ at +3,034 mHa. The λ sweep resolves the transition: the literal square holds $\langle S^2 \rangle \approx 3.0$, with a broad physical variance of ≈ 9 , through $\lambda = 0.1$, then drops to the floor by $\lambda = 0.2$. The projected square cannot leave the contaminated range, and the literal square reaches a singlet only at the full +2.9 Ha cost of the subspace floor.

That both endpoints coincide with the floor of the linear penalty is a mathematical consequence. The operators $(S^2)^2$, $(PS^2P)^2$ and S^2 are all positive semidefinite with the same null space, the exact singlets ($PS^2Pc = 0 \iff \langle c|S^2|c \rangle = 0 \iff S^2c = 0$). Every penalty of this family therefore shares the same +2.9 Ha floor in any compression, and positive semidefiniteness keeps each above the subspace’s unpenalized minimum. This is the noncommutation of the Discussion in operational form. The penalty pulls the ground state toward eigenvectors of the spin operator and the energy pulls it toward eigenvectors of PHP . On a subspace where the two do not commute the compromise has neither identity, whether the penalty is the projected square, held at the wrong value, or the literal square, driven to a singlet 2.9 Ha too high.

The released penalty path does not reach this curve from a cold start. PySCF’s `fix_spin_`, as `solve_sci` connects it, returns `converged = False` on all five (space, shift, max_cycle) arms we ran, at both shifts and at both the silent default max_cycle and max_cycle = 600, each landing 0.27–0.33 Ha above the penalized minimum we certify (Figure 4, panel B). Given our converged penalized ground state as its initial guess, the same kernel returns `converged = True`, overlap 1.000000 and a penalized Rayleigh quotient equal to ours to machine precision. The cold-start failure is therefore an initial-guess problem and not a disagreement about the minimum, and the package’s own kernel certifies the frontier. The one mitigation

the tooling presents as spin-targeting fails without warning when it is switched on. Running the entire configuration-recovery protocol with the penalty active ($ss = 0$, converged solves) reproduces the curve instead of escaping it. At $\lambda = 0.2$ the full loop returns +110.4 mHa at $\langle S^2 \rangle = 3.93$, and at $\lambda = 0.1$ it returns +72.6 mHa at $\langle S^2 \rangle = 4.18$. Both match the fixed-space values to tenths of a mHa, both are worse in energy than the unpenalized +45.5 mHa, and neither is closer to a singlet. The penalty-matched rerun of the published protocol lands where the subspace geometry says it must.

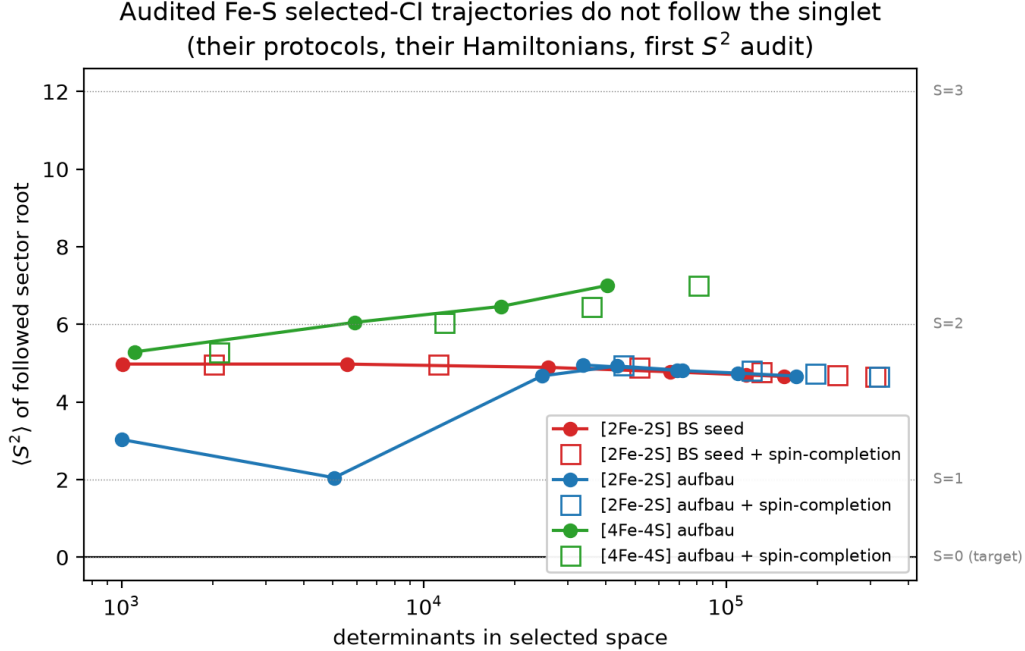


Figure 2: $\langle S^2 \rangle$ of the followed sector root vs subspace size for all audit arms. Open squares: the spin-completed (mitigated) spaces, identical to the unmitigated points. Dotted lines: pure spin eigenvalues.

2.7 The released data archive

The Science Advances data-availability statement points to IBM’s own data archive (https://github.com/jrm874/sqd_data_repository, archived as DOI 10.5281/zenodo.15324153), which contains much more than integrals. It holds the raw hardware measurement outcomes of the flagship experiments ([2Fe-2S]: 2,457,600 shots from December 2023, as 300 batches of 8,192; [4Fe-4S]: 3,163,742 recorded outcomes from April 2024, with device job IDs) and the MO-basis FCIDUMPs those experiments used. It also holds the LUCJ circuit-generation template, a variationally optimized [2Fe-2S] LUCJ circuit with a file described as 58,644,964 noiseless samples drawn from it (Section 2.9), SQD runs on uniform-random samples as null controls, classical HCI ladders, and a configuration-recovery ablation study. Everything in this section is derived from that archive and processed by the released `qiskit-addon-sqd` package, with spin read by the package’s own diagnostic.

The archive’s [2Fe-2S] FCIDUMP is in canonical RHF orbitals. The aufbau determinant in that basis reproduces the tabulated RHF energy (−116.205816 Ha) exactly, with a largest off-diagonal Fock element of 4×10^{-7} , and the tabulated DMRG reference, −116.6056091 Ha, is identical to the near-exact reference of the Li–Chan-basis dumps used in Sections 2.4–2.6. The two are the same active space under a unitary orbital rotation, so our audits and IBM’s experiments address the same spectrum.

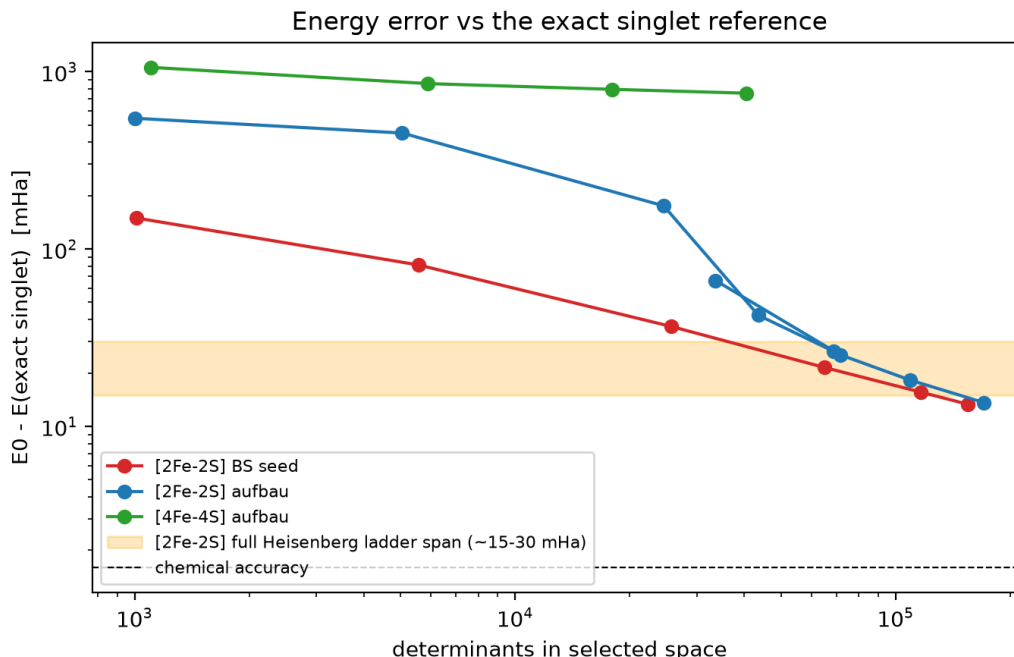


Figure 3: Energy error vs the exact singlet reference. Shaded band: the [2Fe-2S] Heisenberg-ladder span (~ 15 – 30 mHa); dashed line: chemical accuracy.

The archive’s `sqd_on_uniform_distribution` files run the pipeline on uniform random samples as a null control. On [2Fe-2S], the flagship system small enough for uniform sampling to compete, the uniform control matches or beats the hardware samples in every published comparison.

The three published eigenstate files contain two independent evidence streams, a structure we verified row by row before comparing anything and which the audit script asserts. The eigenstate-B hardware file is a strict subset of C’s: all 59 of B’s rows appear bit-identically in C, which carries 7 more. The B and C uniform files are byte-identical, so B and C form a single stream. Eigenstate A shares no rows with either, and its energies differ from C’s at common dimensions. On best energies the hardware trails the uniform control by 1.90 mHa on stream A, where the uniform run used smaller maximum subspaces (1.6×10^7 against 5.6×10^7), and by 12.17 mHa on stream B/C.

The result does not come from picking the best batch. The two protocols share 9 (stream, dimension) conditions, with each hardware file paired with its own uniform null and 5–10 hardware and 10 uniform batches per condition. At all 9, the uniform-control mean energy is lower than the hardware mean: by 4.8–14.9 mHa on stream A and 18.4–26.7 mHa on stream B/C, against per-group sample standard deviations of 3.7–16.3 mHa. Within conditions, whose batches are disjoint shot groups, the uniform advantage is individually significant (two-sided Mann–Whitney, $p < 0.05$) at 7 of the 9; the exceptions are the two smallest-subspace conditions of stream A. We attach no global significance level across conditions. Dimensions within a stream re-process the same released sample pool, so the nine conditions are repeated measures on two independent evidence streams, not nine independent experimental units. The direction (uniform mean $\leq$ hardware mean) holds in all nine, in both streams. An earlier draft counted the B and C files as separate evidence, 13 points in all; the row-level structure audit corrected this to 9. All of these runs, hardware and uniform alike, lie +140 to +184 mHa above the DMRG reference at subspace dimensions up to 5.6×10^7 . In every released [2Fe-2S] comparison available for audit, the hardware samples yield no energy advantage over the corresponding uniform-random control.

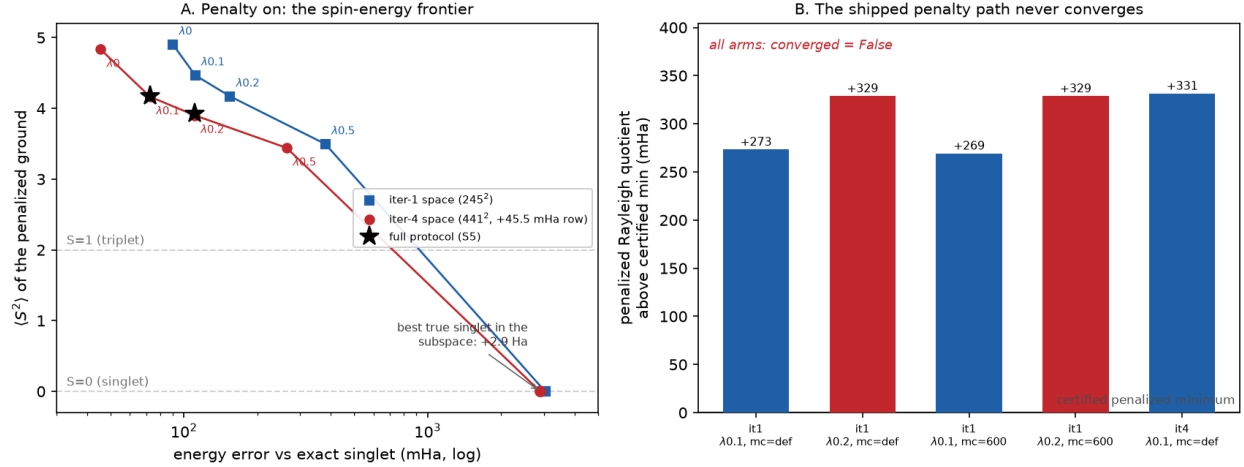


Figure 4: Spin penalty applied to the released pipeline’s [2Fe-2S] subspaces. **A:** $\langle S^2 \rangle$ of the ground state of $H + \lambda S^2$ against its true energy error, for the captured $D = 245^2$ (iteration 1) and $D = 441^2$ (iteration 4, the +45.5 mHa row) subspaces, with markers at each λ , dotted lines at $S(S+1) = 0, 2$, and the full penalized recovery protocol overlaid as stars. The benchmark-energy end ($\langle S^2 \rangle \approx 4.8$) and the singlet end (+2.9 Ha) are ~ 3 Hartree apart, with no competitive point between. **B:** from a cold start the released penalty path does not converge; each `kernel_fixed_space + fix_spin(ss=0)` arm returns `converged=False`, 0.27–0.33 Ha above the certified penalized minimum.

For [4Fe-4S], the archive’s largest published runs (100 batches at subspace dimension $d = 10^8$) reach a best energy of -326.644824 Ha, which is +594.8 mHa above the DMRG reference, and improve by only ~ 53 mHa per decade of subspace growth from $d = 9 \times 10^6$ to 10^8 . The archive’s classical HCI benchmark file reports -326.793854 Ha (+445.8 mHa). At the flagship operating point, on the flagship system, the archive’s quantum-sampled energies therefore trail its classical-selection energies by ~ 149 mHa. Here the hardware does beat uniform sampling, by ~ 1.06 Ha, as it must at 36 orbitals, where uniform 72-bit strings almost never land in a useful part of the space; we measure the correct-sector rate of the raw hardware samples at 5.4×10^{-4} in Section 2.9.

The archive also contains the authors’ N2_6-31G_recovery_effect study. With configuration recovery disabled, its noiseless N₂ SQD errors are 1.1–1.8 Hartree; with recovery enabled, they are 4–7 mHa. The three-orders-of-magnitude difference, at zero noise, is attributable entirely to the classical post-processing. In this ablation, the samples do not carry the accuracy; the classical machinery reconstructs it.

2.8 The released samples through the released pipeline

The results above used our samples and our subspace construction. Here we use IBM’s. We ran the released pipeline with identical settings (`samples_per_batch=500`, 3 batches, `max_dim=500`, seed 17) on every input class the archive makes possible, reading spin with the pipeline’s own diagnostic at every iteration. Table 6 compares input classes at fixed settings; Table 7 then extends the hardware leg to the flagship’s published construction and dimensions.

The hardware leg agrees with the published energetics. Reconstructing shot counts from the archive reproduces the documented protocol exactly ($300 \text{ batches} \times 8,192 = 2,457,600$ shots), and our best energies at subspace dimension 2.5×10^5 are consistent with the published energy–variance rows at the same dimension. We report sector rates as context for what configuration recovery has to accomplish; the pipeline is designed to process out-of-sector samples, so a low rate is not a defect by itself. The raw samples’ correct-particle-sector

Table 6: The released pipeline on every available [2Fe-2S] input class (IBM’s MO basis where the input is IBM’s; identical settings throughout).

input	best E (Ha)	err vs their DMRG (mHa)	their $\langle S^2 \rangle$
10^6 samples of the converged classical benchmark state (Sec. 2.6, Li–Chan basis; err vs same-spectrum exact)	−116.560107	+45.5	4.834
their 2,457,600 raw hardware shots (the flagship experiment’s own data)	−116.358076	+247.5	0.235
10^6 noiseless samples, their circuit template (CCSD t_1+t_2 , their pairs)	−116.376680	+228.9	0.108
uniform draws from their published 58.6M-configuration set (2×10^6 -entry subsample, unit counts)	−116.103827	+501.8	2.008

rate is 0.445%: 99.55% of the flagship experiment’s shots lie outside the physical sector, and the reported subspaces and energies are reconstructed from this small in-sector fraction by the classical recovery loop, consistent with the authors’ recovery ablation (Section 2.7).

The spin–energy trade-off. The first three input classes define a trade-off that no run escapes. Samples of a converged benchmark-protocol state reach benchmark-grade energies (+45.5 mHa) but are strongly spin-contaminated ($\langle S^2 \rangle = 4.83$). The hardware samples and the ideal noiseless samples of the ansatz give states with low scalar $\langle S^2 \rangle = 0.11$ –0.24 that are not competitive in energy (+229 to +248 mHa), because the sampling saturation of Section 2.1 limits the usable subspace. A low mean does not make a clean singlet, and their nonzero variances ($\text{Var}(S^2) = 0.64$ –1.36, Section 2.11) show that these states are spin mixtures as well. No input class approaches both the reference energy and the singlet identity. The mitigation’s only measurable energy gain is on the noncompetitive side. `symmetrize_spin` changes nothing (< 1 nHa) on spin-mixed input but gains 7.7–9.5 mHa on the low- $\langle S^2 \rangle$ noiseless and hardware inputs. It helps only where the returned states already have a low scalar $\langle S^2 \rangle$ (despite their variances) and the energy is not competitive.

2.9 The raw-counts ladder at the published dimensions

The hardware leg at the published construction. The fixed settings above answer the cross-input question. To audit the experiment at its own published operating points, we re-ran the identical pipeline on the same 2,457,600 raw shots with the flagship’s construction. It uses 10 batches per dimension, 5 recovery iterations, and `max_dim` = $N \in \{500, 1000, 2000\}$ strings per spin, giving subspace dimensions $N^2 = 2.5 \times 10^5$, 10^6 and 4×10^6 (Table 7). The last is the largest [2Fe-2S] dimension published below the HPC-scale runs.

Table 7: Raw hardware shots at the published construction (10 batches, 5 recovery iterations, `max_dim` = N): the converged dimension ladder.

<code>max_dim</code> N (dim N^2)	best E (Ha)	err (mHa)	their $\langle S^2 \rangle$	their published best-of-10 (mHa)
500 (2.5×10^5)	−116.368271	+237.3	0.142	+266.3
1000 (10^6)	−116.390277	+215.3	0.108	+238.8
2000 (4×10^6)	−116.438440	+167.2	0.357	+216.7

Our reproduction is, if anything, favorable to the published results. Our best-of-10 beats the published eigenstate-A best-of-10, the only published energetics at these dimensions, by 28.9, 23.5 and 49.5 mHa at matched dimension, and our 10^6 run essentially matches the published 4×10^6 value. Nothing below can therefore be attributed to a handicapped rerun.

The converged $\langle S^2 \rangle$ is not monotone in dimension, and the largest subspace converges to the most spin-contaminated state. The per-iteration record shows why. At $N = 2000$ the first three recovery iterations gain energy while lowering $\langle S^2 \rangle$ (batch values 0.12–0.20 at +189–195 mHa by iteration 3). Iteration 4 then finds a lower-energy, higher-spin basin ($\langle S^2 \rangle = 0.33$ –0.45 at +177–185 mHa), and iteration 5 converges within it (best +167.2 mHa at $\langle S^2 \rangle = 0.357$). The final two iterations gain ~ 21 mHa by leaving the low-spin range, on the flagship’s raw data and by the package’s own diagnostic. This is the spin–energy trade-off operating within a single run. Once the subspace is large enough that the nearly flat noise distribution no longer constrains it, energy-first selection gives up state identity for energy, which is the same behavior that puts the contaminated states at benchmark-grade energies.

The ablations rule out the obvious alternative explanations. A second seed reproduces the 10^6 point to 2.6 mHa and $\Delta\langle S^2 \rangle = 0.001$. Disabling `symmetrize_spin` roughly doubles the converged contamination ($\langle S^2 \rangle = 0.208$ vs 0.108) and costs 6.4 mHa, so on this input class the mitigation does help, although it still does not produce a singlet. An independent run on different hardware lands in the same regime ($\Delta = 6.2$ mHa at 2.5×10^5): the recovery loop’s stochastic draws differ across platforms, but the outcome does not. The best point of the ladder remains +167.2 mHa, two orders of magnitude from chemical accuracy, at $\langle S^2 \rangle = 0.357$. At the experiment’s own published dimensions, on its own data and through its own code, no point combines competitive energy with singlet identity.

[4Fe-4S] hardware samples. We ran the same released pipeline on the 3,163,742 recorded [4Fe-4S] hardware outcomes (measured correct-sector rate 5.4×10^{-4}) at subspace dimension 9×10^4 . With `symmetrize_spin=True` the recovered ground state is a spin-pure triplet, $\langle S^2 \rangle = 2.000$, now confirmed by its second moment ($\text{Var}(S^2) = 3 \times 10^{-6}$, a genuine $S = 1$ eigenstate; cross-instrument table below), at +1,438 mHa from the DMRG reference. With the mitigation off, $\langle S^2 \rangle = 3.00$ at +2,149 mHa. On this system the mitigation does act, unlike its exact no-op on realistic [2Fe-2S] input, and what it does is purify the spin into the wrong sector. The symmetrized subspace provably contains singlet directions, and the variational principle selects the triplet. In every arm of every leg in this paper (HCI-grown, hardware-sampled, template-sampled or set-sampled, with or without mitigation), the named [4Fe-4S] singlet target was never the state returned.

The optimized circuit and the released configuration file. We rebuilt the variationally optimized [2Fe-2S] circuit from its published parameters. The exactly reconstructed state is nearly spin-pure, $\langle S^2 \rangle = 0.000329$, so in this leg the identity failure enters through support truncation, sampling, recovery and projected diagonalization, not through the prepared state itself. (The classically grown state of Section 2.4 involves no circuit at all, and there the problem is intrinsic to selection. The raw hardware and CCSD-template inputs are distinct preparations, not covered by this reconstruction.) The archive’s companion file of “58,644,964 electronic configurations that we sampled” contains 58,644,960 entries, every one of them distinct (the Hartree–Fock string appears exactly once, as row 0). Genuine multinomial draws from the rebuilt state would yield $\sim 5.4 \times 10^6$ distinct configurations at that budget, not 5.86×10^7 . The file therefore cannot be a raw multinomial shot record. It is consistent with the deduplicated union of sampled configurations, the set that defines the ~ 58 M-dimensional subspace ($\sim 24\%$ of the sector), and the raw per-shot record for the optimal-circuit numerics is not public. This is a point about data semantics, not a fault, but it means the sampled subspace and the sampling distribution cannot be audited separately from the release.

The released configuration set as input. The fourth input class (final table row) draws a uniform 2×10^6 -entry subsample of the published 58.6M-configuration set, with unit counts, as the closest public proxy for the input of the optimal-circuit numerics, and runs the identical protocol. At the first recovery iteration $\langle S^2 \rangle$ varies widely from batch to batch (0.97, 3.00 and 4.00 in the mitigated arm). At convergence the mitigated arm settles at $\langle S^2 \rangle = 2.008$ at +501.8 mHa, but this is not a triplet. Its second moment, $\text{Var}(S^2) = 8.0$ ($\langle S^4 \rangle = 12.05$), resolves it into a 2:1 singlet–quintet mixture whose triplet weight is measured below 10^{-4} across every physical spin sector. The third moment, $\langle S^6 \rangle = 72.3$, removes the ambiguity that two moments leave: the alternative with 60% triplet weight, which the first two moments permit, predicts

$\langle S^6 \rangle = 120$ (cross-instrument table below). The mean of 2.00 is therefore a coincidence, one that the measured [4Fe-4S] triplet above ($\text{Var}(S^2) = 3 \times 10^{-6}$) does not share. The unmitigated arm settles at $\langle S^2 \rangle = 1.089$ (strongly spin-contaminated; $\text{Var}(S^2) = 1.5$) at +377.8 mHa.

The mitigation again acts on this input class, moving the mean toward $\langle S^2 \rangle = 2$, but unlike on the [4Fe-4S] hardware leg it does not purify. The mitigated state’s $\text{Var}(S^2) = 8.0$ is broader than the unmitigated arm’s, and here the mitigation also costs energy, ending 124 mHa above the unmitigated arm. This leg lands on neither side of the trade-off: configurations from the curated subspace-defining set, run through the released pipeline, give the wrong spin and energies 378–502 mHa above the reference at once. In no arm of any leg does the singlet appear as the computed answer, meaning a returned low-energy state (within tens of mHa of the reference) that is also a spin singlet (LP-certified $w_0 \rightarrow 1$, $\langle S^2 \rangle \rightarrow 0$). The low- $\langle S^2 \rangle$ arms lie hundreds of mHa high and are themselves broad spin mixtures, while the benchmark-energy arms are strongly spin-contaminated. This is the spin–energy trade-off of Section 2.8 stated as an identity criterion, and no arm satisfies both halves of it.

2.10 The largest published [2Fe-2S] subspace

The largest published dimension. The raw-counts ladder above stops at 4×10^6 because ten-batch recovery runs are limited by desktop resources. The published record extends further, to a single batch at subspace dimension 5.625×10^7 ($= 7500^2$), which is the largest published [2Fe-2S] point and the best published energy for this system anywhere in the archive (−116.421701, +183.9 mHa). We took the spin audit to exactly that dimension, using the idealized best case of the published construction. Outer-product subspaces were built from the top N of the 7,658 released unique half-configurations per spin, ranked by the exact-state marginals of the optimized circuit (nearly spin-pure, $\langle S^2 \rangle = 3.3 \times 10^{-4}$, as measured above), with no sampling noise, no batching and no shot budget. They were diagonalized with the released matrix-free solver (`solve_fermion`, cold start). The only setting changed was `max_cycle`, raised from the silent default of 50 to 400; we disclose it because the default would return an under-converged energy without warning at this dimension. Each rung was certified against our independent PyCI implementation where both can run (energy agreement 0.002–0.023 mHa at $n = 300$ –2000, and bit-identical reruns on fixed hardware and thread count; Table 8).

top- N per spin (dim N^2)	E (Ha)	err (mHa)	their $\langle S^2 \rangle$
300 (9×10^4)	−116.370822	+234.8	3.808
500 (2.5×10^5)	−116.418832	+186.8	3.792
1000 (10^6)	−116.475825	+129.8	3.771
2000 (4×10^6)	−116.525607	+80.0	3.393 / 3.518 (see text)
7500 (5.625×10^7)	−116.591966	+13.6	1.287 delivered → 1.371 certified (see below)

Table 8: The idealized-ranking ladder: top- N exact-marginal subspaces of the released configuration set, diagonalized to convergence with the released solver. The 7500 rung is the largest published [2Fe-2S] dimension.

The flagship rung ran 17.7 hours on a 64-vCPU cloud instance (400 Davidson iterations, final residual 3.5×10^{-4} , energy stable to $< 0.1 \mu\text{Ha}$ per iteration over the last fifty). At the published dimension and in the published product-space form, the idealized exact-marginal construction over the released support converges 170.3 mHa below the published value. This difference measures the combined cost of finite-shot selection, batching, recovery and ranking relative to the best publicly constructible limit of the protocol. Run to convergence, the construction descends to the ~ 13 mHa level reached by the classical selected-CI growth of Section 2.4. The exact determinant selection behind the published point is not public, so this is a same-dimension, same-construction comparison, not a same-subspace one.

The released solver reports $\langle S^2 \rangle = 1.287$ on the state it returns, which is strongly spin-contaminated and incompatible with a singlet. The subspace was ranked by the exact marginals of a state with $\langle S^2 \rangle = 3.3 \times 10^{-4}$, and at 7500 of its 7,658 released half-configurations per spin, the closest public approach to the parent’s own support, the ground state of the selected subspace still does not have the parent’s identity. This is the exact-marginal ranking limit within the released support, the strongest construction publicly buildable for the pipeline, at the published dimension, so the identity failure is not a shot-budget artifact. At a residual of 3.5×10^{-4} in a manifold this dense, however, energy stability does not certify the eigenvector, so we did not stop at 1.287; the certification run below settles the value. Every rung at $n \leq 2000$ is cross-certified to convergence at $\langle S^2 \rangle \geq 3.39$ (run-to-run spread ≤ 0.13), and the flagship rung is certified by the dedicated three-root run below ($\langle S^2 \rangle = 1.371$). No certified or delivered point at any dimension is compatible with the singlet the subspace was built from.

In this manifold $\langle S^2 \rangle$ is not even a stable output. At $N = 2000$, three converged runs of two certified implementations (different machines and thread counts, identical construction, code and integrals) span $22 \mu\text{Ha}$ in energy but 0.126 in $\langle S^2 \rangle$ (3.393 desktop, 3.518 cloud, 3.519 PyCI). At fixed thread count, trajectories reproduce digit for digit across physically different machines, through iteration 9 at the flagship dimension and to the last digit at every certified rung. The only difference among the runs is floating-point summation order, and that alone decides which near-degenerate spin mixture Davidson converges. The energy is a well-conditioned output of the protocol; the spin identity of the state is not.

Certification of the flagship point. Because the delivered flagship vector carries a residual of 3.5×10^{-4} , we ran a dedicated certification (11.3 h at 64 threads): a three-root block eigensolve of the identical $N = 7500$ subspace with the PySCF selected-CI kernel that the released solver wraps. It was warm-started from the delivered vector under three pre-registered gates (bit-identical string sets; warm-vector Rayleigh quotient against the banked energy, measured agreement $0.0 \mu\text{Ha}$; warm $\langle S^2 \rangle$ against banked, $< 10^{-14}$) and converged through a tolerance ladder $10^{-6} \rightarrow 10^{-7} \rightarrow 10^{-8}$, with the rotation-invariant S^2 -Gram measurement repeated at every stage. All three roots meet the solver’s per-root convergence criteria (final residuals $(6.0, 8.4, 9.1) \times 10^{-5}$). The tightest residual is still 21% of the 0.285 mHa gap to the first excited root, so a Davis–Kahan-type bound formally allows ground-pair mixing at the few-percent level, and the certification at this stage rests on stability and invariance rather than on the residual alone. (The stage-4 run below removes this caveat by measurement.) Over the final $100\times$ tolerance tightening, the ground root’s $\langle S^2 \rangle$ moves by 5×10^{-5} and the manifold’s S^2 floor by 6×10^{-4} , three orders of magnitude below the permitted rotation, and the Gram spectrum reported below is invariant under exactly the intra-manifold rotations that the residual leaves undetermined. Only the per-root attribution inherits that sensitivity.

The certified low-energy structure of the flagship subspace is as follows. The lowest root lies at -116.591966 Ha ($+13.643 \text{ mHa}$) with $\langle S^2 \rangle = 1.371$; the first excited state is 0.285 mHa above it, at $\langle S^2 \rangle = 2.357$; and the second is 6.52 mHa up, at $\langle S^2 \rangle = 4.732$. The delivered value had indeed been rotated by the residual, as the conditioning result above predicts. Tightening the residual from 3.5×10^{-4} to 6×10^{-5} moved the energy by $1.9 \mu\text{Ha}$ and $\langle S^2 \rangle$ from 1.287 to 1.371: the spin identity is orders of magnitude more sensitive to convergence than the energy, and the certified state is further from the singlet than the delivered one.

The audit also locates the singlet character that the subspace must contain, since it was built from the exact marginals of a near-singlet state. The invariant S^2 spectrum of the three-root span is $\{0.038, 2.357, 6.066\}$, close to a pure spin ladder $\{0, 2, 6\}$, so a direction with $\langle S^2 \rangle = 0.038$ exists in the span. It is not an eigenstate. It is a coherent superposition of the ground and second-excited eigenstates, with weights 0.78 and 0.22 in the eigenbasis of the compressed 3×3 S^2 block; these are the two roots that bracket it, at 1.371 and 4.732, coupled through S^2 at strength 2.50. (S^2 maps the three-root span partly outside itself, so the eigenvalues of the compressed block are directions within the span, not full spin-sector projector weights.) Its Rayleigh energy lies 1.44 mHa above the variational minimum, and it is not a stationary state of the subspace Hamiltonian. Restricted to the near-degenerate ground pair, the only rotation the spectrum offers at a cost below 0.3 mHa ,

the compressed S^2 floor remains 1.371, stable across the final tolerance stages. The 3×3 block summarizes the result: the subspace contains near-singlet character, but the Hamiltonian’s eigenbasis in this dense manifold does not align with the spin ladder. The eigenstates, which are what energy-ordered selection delivers, mix the block’s lowest and highest spin directions strongly, and none of the three lowest, spanning 6.5 mHa, has $\langle S^2 \rangle$ below 1.37. Tighter convergence does not purify them, and an energy-only benchmark cannot detect the problem.

Stage-4 run: residual, fourth root and second moments. A dedicated stage-4 run (14.8 h at 64 threads; `_gram7500s4.py`) removes the residual caveat by measurement and extends the audit up the spectrum and to second moments. It re-solves the identical subspace, warm-started from the certified block under the same pre-registered gates (ground-root drift $0.0 \mu\text{Ha}$ at every stage), through a second ladder: three roots at tolerance 10^{-10} , then four roots at 10^{-8} and 10^{-10} . The measured ground residual falls from 6.0×10^{-5} to 3.2×10^{-6} , from 21% to 1.1% of the 0.285 mHa ground gap, an 18.6-fold tightening of the ratio the Davis–Kahan caveat concerned. The ground root’s $\langle S^2 \rangle$ is 1.3711 at every stage of both ladders, unchanged in the fourth decimal from the loosest certified solve to the tightest. The invariant three-root spectrum re-derives to $\{0.037, 2.363, 6.072\}$, within 0.007 of the certified values; the upper pair’s per-root attributions shift by ≤ 0.007 , the intra-manifold sensitivity noted above, while the ground root does not move. The rotation the residual formally allowed is now bounded at the 10^{-2} amplitude level and is measurably absent.

One root above the certified manifold, the spectrum continues at +26.19 mHa (12.5 mHa above the ground root) with $\langle S^2 \rangle = 11.59$, predominantly $S = 3$ in character ($S(S+1) = 12$). The four lowest roots of the flagship subspace read $\langle S^2 \rangle = 1.371, 2.363, 4.738, 11.590$, so spin rises with energy (Table 9, Figure 5).

The same run measures $\langle S^4 \rangle$ for each root. The moment kernel was checked in the same process on H_4 FCI states before production (per-eigenstate $|\text{Var}(S^2)| < 10^{-9}$, and an analytic two-state mixture variance reproduced to 10^{-9}). It gives $\text{Var}(S^2) = \langle S^4 \rangle - \langle S^2 \rangle^2 = 6.92, 3.54, 7.41, 4.41$ for the four roots. A spin eigenstate has $\text{Var}(S^2) = 0$, so no low-lying root of this subspace is close to one; the scalar $\langle S^2 \rangle$ values are means of broad spin mixtures, not labels of slightly perturbed eigenstates. The moments also settle a question the compressed Gram block could not, since its eigenvalues are directions within the span rather than sector weights. A two-moment sector solve asks whether the pair $(\langle S^2 \rangle, \langle S^4 \rangle)$ admits nonnegative weights on $S \in \{0, 1, 2\}$. It is feasible only for root 1 (weights 0.18/0.63/0.18). For the ground root it returns $(0.82, -0.07, 0.25)$, which is infeasible, so the flagship ground state, the object closest to a singlet that this construction produces (mean $\langle S^2 \rangle = 1.371$), provably carries spin character beyond $S = 2$. The decomposition is identical to four decimals across all four checkpoints of the run. The low-energy eigenstates of the flagship subspace are therefore not near-singlets, not near any spin eigenstate, and not even confined to the low-spin sectors that their $\langle S^2 \rangle$ means suggest. The two outcomes of the sector solve are not symmetric. A negative weight is a certificate that no $S \leq 2$ distribution reproduces the moments, whereas a feasible nonnegative reading (root 1 here) shows only that an $S \leq 2$ decomposition is possible, not that the support is confined there. At 5.625×10^7 determinants the third moment that would settle the feasible cases requires a terabyte-scale S^+ -ladder pass that we did not run. The tractable pipeline states of Table 10, of dimension $\leq 500^2$, are carried to $\langle S^6 \rangle$, which resolves the corresponding ambiguity by measurement, while the three infeasible rows above stand as certificates.

Independent implementation at the flagship dimension. Every flagship number above, the solve and its certification alike, comes from one kernel: PySCF’s `kernel_fixed_space`, the matrix-free selected-CI engine that the released solver wraps. Earlier drafts listed that dependence on a single kernel as a limitation. We removed it with a second implementation that shares no Hamiltonian-construction or solver code with the first. PyCI built the Hamiltonian over the identical 5.625×10^7 -determinant subspace explicitly, as a 1.6 TB on-disk sparse triangle whose nonzero count, 127,975,471,122, matches the closed-form combinatorial prediction exactly: twice, in two independent builds on two fresh cloud instances. The eigenproblem was solved by our own block Davidson over our own threaded matrix–vector kernels (Methods).

The certified stage-4 vectors are eigenvectors of the independently built operator at machine precision:

Table 9: Flagship-dimension subspace (5.625×10^7 determinants, reconstructed here), four lowest roots (stage-4 run, $\text{tol} = 10^{-10}$; checkpoint `gram7500_s4ck3.npz`): energy above the ground root, first two spin moments, variance, and the $\{S = 0, 1, 2\}$ two-moment sector solve. $\text{Var}(S^2) = 0$ marks a spin eigenstate; a negative weight certifies support outside $S \leq 2$. [†]Root 3 reached the final stage’s iteration cap (residual 3.3×10^{-5}) and is reported as the returned vector, not a converged eigenpair (Methods); its $\langle S^2 \rangle > 6$ infeasibility certificate holds for any vector, while its energy and higher moments are those of that vector.

root	ΔE (mHa)	$\langle S^2 \rangle$	$\langle S^4 \rangle$	$\text{Var}(S^2)$	$\{S \leq 2\}$ weights $\rightarrow$ reading
0 (ground)	0	1.371	8.795	6.92	(0.82, -0.07, 0.25) infeasible $\rightarrow$ requires $S \geq 3$ support
1	+0.285	2.363	9.127	3.54	(0.18, 0.63, 0.18) admits $S \leq 2$ (feasible, not unique)
2	+6.52	4.738	29.857	7.41	(0.33, -0.18, 0.85) infeasible $\rightarrow$ requires $S \geq 3$ support
3 [†]	+12.54	11.590	138.740	4.41	$\langle S^2 \rangle > 6$ infeasible $\rightarrow$ predominantly $S = 3$

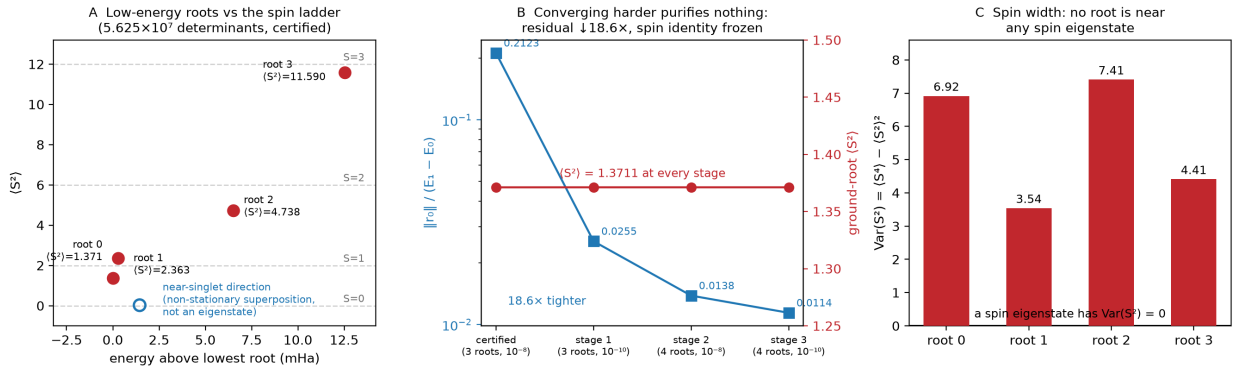


Figure 5: The certified flagship-dimension subspace (5.625×10^7 determinants, reconstructed here), four lowest roots. **A:** $\langle S^2 \rangle$ vs energy above the ground root, with dotted lines at the pure-spin values $S(S+1) = 0, 2, 6, 12$; the near-singlet direction (open circle) is a non-stationary superposition 1.4 mHa above the minimum, not an eigenstate. **B:** across the stage-3/stage-4 certification the ground residual-to-gap ratio falls 18.6-fold ($0.2123 \rightarrow 0.0114$) while the ground-root $\langle S^2 \rangle$ holds at 1.3711. **C:** $\text{Var}(S^2)$ per root; a spin eigenstate has $\text{Var}(S^2) = 0$, so no low-lying root is near one. Root 3 ($\langle S^2 \rangle = 11.59$) is the iteration-capped Ritz vector (Methods), the high-residual point in B, plotted as returned.

their Rayleigh quotients match the certified energies to $|\Delta| = 1.4 \times 10^{-14}$, 2.0×10^{-13} and 1.3×10^{-13} Ha. This establishes the independence of the kernel without any solver at all. Perturbed by a deliberate 10^{-5} -scale jitter per component (initial block residual 2.0×10^{-1} , energies ≈ 14 mHa off), the independent solver re-converges all three roots in 38 iterations. The energies return to within $(0.4, 0.6, 0.8) \times 10^{-9}$ Ha of the certified values, and the spin ladder to $\langle S^2 \rangle = 1.371065, 2.362748, 4.737714$, within 5.2×10^{-5} , 7.5×10^{-4} and 3.0×10^{-4} of the certified ladder. The independent solve stopped at a maximum residual of 1.25×10^{-5} (4.4% of the 0.285 mHa ground gap), at a pre-registered wall-clock budget; the slow tail of the diagonal preconditioner against the near-degenerate pair is documented in the run logs. Davis–Kahan-type bounds therefore allow intra-pair rotation up to $\sin \theta \approx 0.044$, which bounds the eigenvalue error by 5.5×10^{-7} Ha and the $\langle S^2 \rangle$ mixing by 1.9×10^{-3} . Every observed deviation lies inside its bound, the energies by a factor of ~ 700 and the spin by 2.5. At the flagship dimension, two implementations that share no code agree on the manifold, its energies and its spin identity to the precision either can state, which extends to the published dimension the two-solver agreement maintained at every rung $n \leq 2000$.

2.11 Higher spin moments and numerically certified sector bounds

Measurement check on the pipeline states. We first compared our independent S^2 -Gram measurement (Section 2.3) with the released `SCIState.spin_square()` on the final states of the [2Fe-2S] hardware, template and set-draw legs and of the [4Fe-4S] hardware leg, with mitigation on and off in each (eight arms in all). The two agree to better than 3×10^{-14} , so they compute the same observable on the same states. We then extend the moment ladder two steps past the mean, to $\langle S^6 \rangle$ and $\langle S^8 \rangle$, using the same $M_s = 0$ S^+ -ladder identities. These were checked on H4 FCI through the fourth moment and on pure high-spin $S = 3/4/5$ reference states (built as the top eigenvector of PySCF’s own `contract_ss` S^2 operator, independent of the S^+ kernel), which exercise the third and fourth raising steps that the H4 check cannot reach (Methods). Each state’s measured moments are then passed to a linear program over the full physical sector grid ($S = 0 \dots 5$ for [2Fe-2S], $0 \dots 9$ for [4Fe-4S]), which returns moment-constrained sector bounds: numerically certified minimum and maximum weights for every sector. Here “certified” means bounds on the feasible set at an LP tolerance of 10^{-7} -scale, not an analytic proof. This replaces the two-moment sector assumption with measurement.

A nonnegative reading from $\langle S^2 \rangle$ and $\langle S^4 \rangle$ alone is feasible but not unique. For the set-draw state, a $\{0, 1, 3\}$ mixture with 60% triplet weight reproduces the first two moments exactly, and the two-moment feasible set allows a triplet weight as high as 0.86. The third moment separates the cases: that alternative predicts $\langle S^6 \rangle = 120$, the state measures $\langle S^6 \rangle = 72.30$, and adding this one moment collapses the certified triplet ceiling to 8.4×10^{-5} (the fourth moment, $\langle S^8 \rangle = 434.1$, tightens it to 3.0×10^{-5}). The full linear program over all physical sectors then fixes the decomposition at $w_0 = 0.665$ (singlet) and $w_2 = 0.335$ (quintet), with the triplet weight below 4×10^{-5} and the combined $S \geq 3$ weight below 2×10^{-5} . The 2:1 singlet–quintet reading is therefore measured in full, not inferred from the mean (Table 10).

Table 10: Moments and numerically certified sector bounds of the eight final pipeline states. Two arms share $\langle S^2 \rangle \approx 2.00$; the higher moments and the linear-program triplet weight w_1 (numerically certified over all physical sectors) separate the genuine triplet from the singlet–quintet mixture.

leg (symmetrize_spin)	$\langle S^2 \rangle$	$\langle S^4 \rangle$	$\langle S^6 \rangle$	$\text{Var}(S^2)$	w_1 (LP, all S)	reading
[2Fe-2S] hardware, on	0.235	1.412	8.50	1.357	$< 10^{-4}$	contaminated
[2Fe-2S] hardware, off	0.231	1.205	6.95	1.152	0.024	contaminated
[2Fe-2S] template, on	0.108	0.650	3.96	0.638	$< 2 \times 10^{-4}$	contaminated
[2Fe-2S] template, off	0.123	0.705	4.29	0.690	0.006	contaminated
[2Fe-2S] set-draw, on	2.008	12.047	72.30	8.016	$< 10^{-4}$	singlet–quintet mix, no triplet
[2Fe-2S] set-draw, off	1.089	2.693	9.92	1.507	0.498	contaminated
[4Fe-4S] hardware, on	2.000	4.000	8.00	3×10^{-6}	1.000	triplet eigenstate, to tol. ($S = 1$)
[4Fe-4S] hardware, off	3.003	18.048	144.73	9.028	0.450	broad mixture, $\geq 5\%$ at $S \geq 3$

Two arms converge to $\langle S^2 \rangle \approx 2.00$ and each could be reported as “the triplet” on the scalar mean alone. The higher moments separate them, and the linear program makes the separation numerically certified. The [4Fe-4S] hardware state with mitigation on is a triplet eigenstate to measured tolerance: $\text{Var}(S^2) = 3 \times 10^{-6}$, and the LP fixes its triplet weight at $w_1 = 1.000$ over all sectors $S \leq 9$ (the variance alone gives a Chebyshev cap of $\text{Var}/4 \approx 7 \times 10^{-7}$ on non-triplet weight). There the mitigation does purify the state, from $\text{Var}(S^2) = 9.0$ unmitigated, but into the wrong sector. The [2Fe-2S] set-draw state with mitigation on is not a triplet at all. Its third moment bounds the triplet weight below 10^{-4} across every physical sector (LP ceiling 8.4×10^{-5}), so the 2:1 singlet–quintet reading is measured rather than assumed: the mean of 2.0 is a coincidence, and the 60%-triplet alternative that the first two moments permit is excluded by the third. The other five [2Fe-2S] arms are broad low-spin mixtures ($\text{Var}(S^2) = 0.6$ – 1.5 , LP triplet weight ≤ 0.5), and the unmitigated [4Fe-4S] arm is a broad high-spin mixture with a certified $\geq 5\%$ weight in sectors $S \geq 3$. Across the eight output states,

a scalar $\langle S^2 \rangle$ labels none of them, and exactly one of the eight is a spin eigenstate to measured tolerance, in the wrong sector. (The full $\langle S^8 \rangle$ per arm and the complete per-sector LP bounds are archived in `s6moments.npz`.)

We also stress-tested the bounds against their tolerance boxes. Each box is a per-moment uncertainty interval ($10^{-7} \cdot \max(1, |m_k|)$), so the LP propagates uncertainty, and inflating every box 10 $\times$ and 100 $\times$ moves the set-draw triplet ceiling only from 3.1×10^{-5} to 3.7×10^{-5} and 1.0×10^{-4} , confines its singlet weight to $[0.6653, 0.6654]$, and leaves the [4Fe-4S] triplet floor above 0.99998. The sector readings are properties of the measured moments, not of tight tolerances (`_s6stress.py`, `s6stress.npz`; all eight states, factors up to 1000 $\times$).

3 Discussion

Both positions in the dispute treat “the energy at subspace size D ” as the benchmark quantity. On the iron–sulfur systems that quantity does not identify a state. The low-energy manifold is a dense spin ladder, truncated-CI selection has a measured and persistent bias toward compact high-spin mixtures, and both quantum-sampled and classically selected subspaces converge to unidentified spin-contaminated mixtures. The critique’s framing in terms of good and bad initial guesses does not apply, since both guesses reach the same state, and the flagship protocol does not produce the named singlet in any audited realization, with or without its mitigation. At the accuracy scale where the two sides disagree, a few mHa to a few tens of mHa, the spin-identity error is the same size as the disagreement, so the dispute cannot be settled on energies alone. It can be settled by measuring the state.

What each audited source claimed, and what we measured. Because this critique concerns published claims, its scope should be explicit. Table 11 places each audited source’s own words beside the state diagnostic it reported and what this audit establishes, so that it is clear we do not contest a stronger statement than was made. The variational upper bound is granted: it is a true statement about an energy (see “What the energy still means,” below). What is not met is the state identity that each source either names or leaves unmeasured. The one source that names a spin target, the flagship (“singlet states . . . eigenfunctions of total spin with eigenvalue 0”), reports an energy but never an $\langle S^2 \rangle$. The spin-handling mechanisms the sources invoke (“closure under spin inversion,” and, for the flagship, the stated soft-constraint penalty at $\lambda = 0.2$) are the ones measured in Section 2.6. Closure leaves the ground state unchanged, and the stated penalty, in its released linear form and its stated squared form alike, yields no competitive singlet and reaches one at best at the subspace’s +2.9 Ha floor. The full-scale successor, which carries the program beyond what we could audit, names no spin target and reports no spin diagnostic.

The alignment is deliberately narrow. It does not claim that no SQD execution could ever return the named singlet, or that the unpublished flagship eigenvector was certainly contaminated. It establishes that on every execution this audit could reach (the trajectories, the released pipeline, the released hardware shots and the reconstructed flagship-dimension subspace), the returned state carries a spin width that the reported energies did not disclose.

The structural mechanism. The pattern has a single cause. Determinant selection is a projector P that is not S^2 -invariant, so although $[H, S^2] = 0$ exactly, the compressed operators need not commute: $[PHP, PS^2P] \neq 0$, and the eigenbasis of PHP need not align with any spin sector. Every spin-identity finding in this audit follows from that noncommutation. The compactness bias arises because the ground state of PHP optimizes the energy in a space where spin sectors mix. Spin-inversion closure fails because it constrains the support of P without making PS^2P commute with PHP (Section 2.6 measures the consequence). And the near-singlet direction in the flagship-dimension manifold is not an eigenstate: Section 2.10 exhibits the compressed 3×3 blocks directly, and its second-moment audit finds $\text{Var}(S^2) \geq 3.5$ for every low root, which is spin width rather than a spin label. (The sampling saturation of Section 2.1 is a separate mechanism, distribution concentration, not broken commutation, and it limits support recovery rather than state identity.)

The noncommutation also separates three conditions that benchmark practice often conflates: (i) the target state is representable in the subspace; (ii) target character is present within the low-energy manifold; and (iii) the eigensolver returns the target as an eigenstate. Only the third supports a named-state energy. Spin-inversion completion achieves the first (Section 2.6), and the flagship Gram audit exhibits the second, a near-singlet direction inside the certified low-energy span (Section 2.10), but neither delivers the third. Removing the noncommutation requires an S^2 -invariant construction, such as spin-adapted (CSF) bases or S^2 -projected selection, for which $[PHP, PS^2P] = 0$ by construction. A solver-side penalty is not such a construction. Adding any function of PS^2P , whether the linear λPS^2P or the flagship’s stated squared $\lambda [PS^2P]^2$, leaves P unchanged and commutes with PS^2P , so the noncommutation with the projected energy is unaltered. The penalty reshapes the spectrum of PHP and biases its ground state toward whatever low-spin directions the selected space already represents, which is why it can reach a competitive singlet only after selection has placed one in the subspace (Section 2.6 finds both forms reaching the same floor). The recommendations below follow from this.

What the energy still means. Spin contamination does not erase the variational meaning of the energy; it erases the state-specific meaning assigned to it. A contaminated $M_s = 0$ root remains a variational upper bound on the global ground-state energy and can be quoted as one. What it cannot support is identification with the named singlet, whether of its energy, its wavefunction or any spin-ladder observable read from it.

The full-scale successor. While this audit was under way, the flagship program scaled again. A RIKEN–IBM collaboration (arXiv:2511.00224, November 2025) reports closed-loop SQD on [2Fe-2S] (50e, 36o, a larger active space than the benchmark audited here) and on [4Fe-4S] (54e, 36o) at subspace dimensions up to 10^9 – 10^{10} on Fugaku. Its [4Fe-4S] DMRG reference, -327.239 Ha, matches the benchmark dump’s to the milli-Hartree, so on this system the successor and our audit address the same spectrum. It improves the best [4Fe-4S] energy to -326.912 Ha at $D = 10^{10}$ and extrapolates in the energy–variance plane to -327.118 ± 0.02 Ha, a value its own text notes “deviates by about $0.12 E_h$ ” from the DMRG reference.

Three features of the successor, each verifiable from its full text, place it within the scope of this critique. It reports no spin measurement of any kind: “singlet” and “triplet” each occur exactly once in its full text, inside the title of a cited reference, and no $\langle S^2 \rangle$ or spin diagnostic appears, so the computed state is identified by energy alone. Its spin mitigation is the mechanism measured here: mirror symmetrization of the sampled α/β half-configurations, with the Hartree–Fock string “manually added” when sampling misses it, which is the same product-space completion whose ground manifold contains no singlet at any scale we audited (Sections 2.6, 2.8). And its classical baselines are CCSD and the DMRG reference; no selected-CI baseline at matched subspace dimension is run, so the matched-dimension question of Section 2.1 is unexamined at 10^{10} . The successor’s central accuracy argument, extrapolation to zero energy variance, does not substitute for state identification. Extrapolating the energies of unidentified states targets whichever eigenstate the trajectory approaches, and a terminal 0.12 Ha gap from DMRG is exactly the signature an energy-only protocol cannot diagnose: slow convergence toward the singlet and faithful convergence toward a different, unidentified eigenstate look the same without a spin measurement. One sparse matrix–vector product per reported state would settle it.

This inheritance argument has definite limits. It is an argument from shared mechanism and shared data handling, not a spin measurement of the successor’s own states, which at 10^9 – 10^{10} remain beyond our audit (Limitations item 1), and the paper does not rest on it. The main findings are the fully measured lower-dimension results on IBM’s own integrals, subspaces, hardware shots and released pipeline (Sections 2.7–2.11), which identify the failure directly whether or not the successor is ever audited at scale.

Where quantum sampling could still matter. Our results settle the matched-dimension question in every regime we could reach exactly (equilibrium, stretched and diradical, with three certified implementations and three seeds) and add the identity failure at flagship scale. The remaining candidate value of quantum sampling is not accuracy per determinant but escape from selection hysteresis, which is real (Section 2.2): samples from a state on the correct potential-energy sheet cannot be trapped by a wrong-state seed. That value is

realizable only if the circuit prepares approximately the right state, and the flagship initialization measurably does not do so at exactly the near-degeneracies where escape matters: CCSD-seeded LUCJ at the diradical gives 181 unique determinants and +64 mHa, with the wrong-state bias inherited from CCSD. A claim of quantum advantage for SQD therefore requires, at minimum, (i) a demonstrated preparation whose sample distribution favors the named target state where classical selection demonstrably follows a different one, and (ii) spin and other symmetry identification of every reported state. Neither exists in the current literature.

Falsification criteria. The claims are falsifiable: they fail if anyone can exhibit any of the following. (1) A quasi-degenerate ground manifold (Methods: $\Delta E < 2 \mu\text{Ha}$) of a spin-completed subspace of these benchmark systems whose S^2 -Gram floor is < 1 (target 0), using the Gram measurement provided. Rotations across gaps far larger than the degeneracy window are superpositions, not eigenstates, and do not qualify. The certified flagship manifold of Section 2.10 illustrates the distinction, with a ground-pair floor of 1.371 and a $\langle S^2 \rangle = 0.038$ direction that exists only as a non-stationary mixture across a 6.5 mHa gap. (2) A quantum-sampled subspace at matched determinant count whose spin-identified ground-state energy beats HCI or CIPSI on any system here. (3) A run of the released pipeline, on any sample source in IBM’s archive, that escapes the spin–energy trade-off ($\langle S^2 \rangle < 1$ and error < 50 mHa on [2Fe-2S]), with or without a solver-side spin penalty. We have measured that a solver-side penalty does not escape it: with the released linear `fix_spin_` at and above its default, and with the flagship’s stated squared constraint at $\lambda = 0.2$, the benchmark-energy subspace reaches a singlet only 2.9 Ha up (Section 2.6). Escaping through spin-aware selection would confirm our recommendation rather than refute our claim. The harnesses are in the supplementary archive.

Recommendations. (1) Every selected-CI-family benchmark on open-shell systems, classical or quantum-sampled, should report the determinant-space $\langle S^2 \rangle$ for every state whose energy is reported, and within quasi-degenerate manifolds the invariant S^2 -block eigenvalues. The cost is negligible. Spin moments are the first identity certificate, not the complete one: in a dense exchange ladder a spin-pure root can still be the wrong root of its sector. A full certificate therefore adds the remaining symmetries and target diagnostics that the named state defines (spatial irreducible representation, particle number, and overlap or root continuity against a declared target), and an energy becomes state-specific only when the returned eigenvector passes the certificate declared in advance. (2) Spin control should be applied during selection, not only at diagonalization. Subspace completion alone is measurably insufficient (Section 2.6), and so is a spin penalty applied afterward to a marginal-product subspace. In the released linear form or the flagship’s stated squared form, such a penalty reaches a singlet only 2.9 Ha above the reference because the sampled subspace contains no competitive singlet (Section 2.6). The fix has to act at the level of spin-adapted CSF bases or S^2 -projected selection that keeps the recoupling determinants; a `fix_spin_`-style penalty can then target the singlet within a subspace that contains one. (3) Matched-dimension discipline, in which every quoted D is the dimension actually diagonalized, should be standard in QSCI and SQD comparisons. It is the first-order cost axis but not the whole cost model: connectivity, product-space structure and solver internals also set wall-clock time and memory, which is why our Methods report iteration counts, wall times and peak memory for each leg. (4) Iron–sulfur claims of results “beyond exact” diagonalization should be restated as energies of identified states, or explicitly rescoped as probe-regime results.

Table 11: Claim alignment across audited sources. Each row pairs a source’s own stated claim (verbatim in quotation marks) with the state diagnostic it reported and what this audit establishes. No audited source that names a spin target reports a spin diagnostic for its computed states; the invoked spin-handling mechanisms, closure under spin inversion and the flagship’s stated soft-constraint penalty, are measured in Section 2.6.

Source	Stated claim (verbatim in quotes)	Names a spin target?	State diagnostic reported	What this audit establishes
IBM flagship , arXiv:2405.05068 (Sci. Adv. 2025)	“upper bounds for the ground-state energy” and “sparse approximations to the ground-state wavefunctions” for systems “beyond sizes amenable to exact diagonalization”	Yes : computed “in the $S_z = 0$ and $S^2 = 0$ subspace,” targeting “singlet states ... eigenfunctions of total spin with eigenvalue 0”	Energy only; no $\langle S^2 \rangle$. Spin handled by “approximate restoration of the S^2 symmetry ... closure under spin inversion” plus “a soft constraint in the eigenstate solver ... to mitigate spin contamination” $(H + \lambda[S^2 - s(s + 1)]^2, \lambda = 0.2)$	The upper bound stands as a statement about an energy; but the audited trajectories and the reconstructed flagship subspace carry $\langle S^2 \rangle$ from 1.37 upward, never the singlet’s 0; spin-inversion closure leaves the ground state unchanged; and the stated constraint, run at its stated form and strength on the benchmark-energy subspace, yields no competitive singlet in either compression (the projected square holds $\langle S^2 \rangle = 3.01$ at +528 mHa; the literal square $P(S^2)^2P$ reaches a singlet only at the +2.9 Ha floor) (Secs. 2.6, 2.10). The reported energy is not identified with the singlet.
IBM data archive , zenodo.15324153	Released hardware shots, noiseless-template samples, uniform controls, and a recovery ablation for the demonstrations above	N/A (released data)	Energies; no per-state $\langle S^2 \rangle$	Direct archive result: the archive’s own inputs, run through the shipped pipeline, show the spin–energy trade-off: benchmark-grade energy only at $\langle S^2 \rangle = 4.8$, or low scalar $\langle S^2 \rangle$ only at +229–248 mHa (where the raw hardware shots land), never both (Secs. 2.7–2.9); the authors’ own recovery ablation attributes the accuracy to the classical post-processing (Sec. 2.7).
Shipped software , qiskit-addon-sqd (tested 0.3.0–0.12.1)	Diagonalizes sampled subspaces; offers a <code>symmetrize_spin</code> closure and a <code>fix_spin_</code> penalty; exposes <code>SCISState.spin_square()</code>	Optional: spin symmetrization and penalty are available but off by default	$\langle S^2 \rangle$ obtainable via <code>spin_square()</code> but not invoked on the default path; the solver can return <code>converged=False</code> without raising	Version-tested behavior: <code>symmetrize_spin</code> changes the result by <1 nHa on realistic spin-mixed input (no effect on the inputs where it would be needed), the default path measures no spin, and the penalty is off by default (Secs. 2.6,

4 Methods

Terminology: four senses of “certified.” This paper uses the word in four distinct senses, each clear from context and none implying the others. A result is *residual-certified* when an eigenpair met a stated solver tolerance; it is quoted with its measured residual and the local gap that residual must be compared against. It is *cross-kernel reproduced* when an independent implementation re-derived it (the PyCI-built operator and in-house eigensolver of Section 2.10). It is *numerically certified* when it is an LP minimum or maximum over the moment-feasible set at stated tolerances (defined at first use in the flagship-leg paragraph below), not an analytic bound. And it is *artifact-verified* when `_paperaudit.py` re-derives the manuscript statement from the archived outputs, which establishes internal consistency only (see “Automated claim verification”).

Certification chain. All Hamiltonians are read from the public Li–Chan FCIDUMPs ([2Fe-2S]: 20 orbitals, 30 electrons; [4Fe-4S]: 36 orbitals, 54 electrons). Our Slater–Condon engine, PySCF, and PyCI agree to μHa on diazene FCI cross-checks; production iron–sulfur ladders use PyCI (`add_hci` with ε -ladder factor $10^{-0.1}$, `sparse_op.solve` to 10^{-9}). Our HCI implementation independently reproduces the critique’s published N_2 HCI curve. PySCF’s own SCI independently reproduces the diazene triplet trap. The [2Fe-2S] 170k-determinant space re-solves to the recorded energy exactly on reload (stage-1 check of the as-shipped test).

Spin audit. $\langle S^2 \rangle$ and S^2 -Gram blocks computed as in Section 2.3, with the diazene gauge gate (known triplet/singlet/singlet labels; required diagonal (2, 0, 0), off-diagonals $< 10^{-6}$) run before every production audit. Quasi-degenerate clusters defined by $\Delta E < 2 \times 10^{-6}$ Ha. That convention sits on a measured plateau (`_thrscan.py`, `thrscan.npz`): re-deriving every cluster partition from the archived root energies (49 stored root vectors (138 roots) across the trajectory, completion, completion-mechanism, and flagship records) reproduces the production partition exactly at $0.2 \mu\text{Ha}$, at $2 \mu\text{Ha}$, and under a residual-adaptive criterion (roots cluster when their gap falls below the per-root eigenvalue uncertainty: the solve tolerance where residuals are not archived, the Kato–Temple-style r^2/gap bound where they are). The partition does not move because it is set by a separation margin: the largest split inside any production cluster is 1.9×10^{-12} Ha, the smallest gap between distinct clusters $4.8 \mu\text{Ha}$, a 2.6×10^6 -fold window around the convention. Only at $20 \mu\text{Ha}$ does exactly one assignment change: the two lowest roots of the [2Fe-2S] aufbau trajectory checkpoint at $D = 5,053$ (gap $4.8 \mu\text{Ha}$; per-root $\langle S^2 \rangle = 2.05$ and 2.07 , both mid-mixture, the wandering point of Section 2.4) merge into a cluster the paper never reported as one; no completion pair, no flagship manifold, and no reported block spectrum is affected. Spin completion implemented as full spin-inversion closure of the determinant set, re-solved with identical solver settings.

Sampling and matched dimension. Sampling arms draw multinomial shots from $|c|^2$ of the stated vector (exact-distribution arms) or from LUCJ circuit simulation (ffsim, CCSD t_2 initialization, with determinant-mapping self-tests). Classical growth uses $\times 1.5$ incremental selection with full re-solve so that every reported D is the diagonalized dimension. Diazene comparisons use $B = 2 \times 10^6$ shots and three RNG seeds.

As-shipped pipeline. We call `diagonalize_fermionic_hamiltonian` from `qiskit-addon-sqd` (installed release) with `samples_per_batch=500`, `norb=20`, `nelec=(15,15)`, `num_batches=3`, `max_iterations=4`, `symmetrize_spin` in `{True,False}`, `max_dim=500`, `seed=17`; bitstrings packed $[\beta \dots, \alpha \dots]$ per its convention; 10^6 multinomial shots from the $D = 170,448$ aufbau vector; energies reported with `ECORE = 0` (per the FCIDUMP); $\langle S^2 \rangle$ from `SCIState.spin_square()`. Per-batch trajectories in `sqdship_symm1.npz` and `sqdship_symm0.npz`. The published-construction ladder of Section 2.9 uses the same driver on the reconstructed 2,457,600-shot hardware BitArray with `samples_per_batch = 3N`, `num_batches = 10`, `max_iterations = 5`, `max_dim = N` in `{500,1000,2000}`, `seed 17` (seed 43 and `symmetrize_spin=False` ablations at $N = 1000$; an independent same-configuration run of $N = 500$ on second hardware), energies vs the same reference; per-batch trajectories in `sqdraw_n{N}_{arm}_s{seed}.npz`.

Penalty-on audit (Section 2.6). `_sqdpen.py`. We first captured the shipped off-arm subspaces by

rerunning `diagonalize_fermionic_hamiltonian` on the 10^6 samples of the converged HCI vector at the Section 2.6 knobs (`samples_per_batch=500`, 3 batches, `max_dim=500`, 4 iterations, seed 17), saving every iteration’s product-space CI strings; the best space ($D = 441^2$) reproduces the published -116.560107 Ha / $\langle S^2 \rangle = 4.834$ and the iteration-1 space ($D = 245^2$) reproduces -116.515503 Ha / 4.903. On each fixed space we solved the ground state of $H + \lambda S^2$ by Lanczos (`scipy.sparse.linalg.eigsh`, $k = 1$, `ncv=64`) whose matrix–vector products call PySCF’s own `selected_ci.contract_2e` (for H , via `absorb_h1e`) and `contract_ss` (for the projected S^2), the exact operators `fix_spin_` penalizes and `spin_square` measures, over the penalty grid $\lambda \in \{0, 0.1, 0.2, 0.5, 1, 2, 5, 20\}$, warm-started by continuation, reporting the true energy $\langle H \rangle$ (from the RDMs), $\langle S^2 \rangle$, and $\text{Var}(S^2)$ of each converged root with its explicit residual ($\leq 9 \times 10^{-8}$). At $\lambda = 0$ the eigensolver reproduces the shipped record and agrees with an independent PyCI diagonalization of the identical determinant space to $0.00 \mu\text{Ha}$. The singlet-sector endpoint (large λ ; $\langle S^2 \rangle = \text{Var}(S^2) = 0.0000$, an exact singlet by positive-semidefiniteness) is the lowest-energy exact singlet the subspace contains, corroborated by the lowest closed-shell determinant’s diagonal energy as an independent single-determinant upper bound ($+3.05$ Ha; `_sqdpen_diagprobe.py`). We then reran their solver on the same fixed spaces exactly as `solve_sci` wires it (`kernel_fixed_space` with `fix_spin_(ss=0, shift \in \{0.1, 0.2\})`), at the silent default `max_cycle` and at `max_cycle=600`. Every one of the five arms returned `converged=False`, $0.27\text{--}0.33$ Ha above the penalized minimum. Seeded instead with our converged penalized vector, the same kernel returns `converged=True` at overlap 1.000000 and our exact penalized Rayleigh quotient (`_sqdpen_s3b.py`, `sqdpen_s3b.npz`). Finally we reran the entire configuration-recovery protocol (same knobs and seed) with the penalty active ($ss = 0$, converged eigensolver solves) at $\lambda = 0.1$ and 0.2 . Two companion runs solve the flagship’s stated squared constraint, $H + \lambda[S^2 - s(s+1)]^2$ at $s = 0$ and $\lambda = 0.2$, on the same captured $D = 441^2$ space, in the two ways that operator compresses onto a fixed subspace (PySCF’s `fix_spin_` implements only the linear shift, so both require a dedicated operator). `_sqdpen_sq.py` applies `contract_ss` twice, the projected square $(PS^2P)^2$; `_sqdpen_true.py` builds the literal square $P(S^2)^2P$ from the exact $M_s = 0$ spin ladder (sparse S^+ operators mapping P to the M_s+1 and M_s+2 product spaces, so that $P(S^2)^2Pc = R_1^T(R_2^T S_+^2 c + 2S_+ c)$), gauged operator-for-operator against PySCF’s `contract_ss` and a dense projected S^4 to machine precision (the G0 gate) before the run. Both use Lanczos as above, warm-started, with the $\lambda = 0$ anchor certified against the shipped record by the same gates. Outputs `sqdpen_frontier.npz`, `sqdpen_theirs.npz`, `sqdpen_spaces.npz`, `sqdpen_loop_l{10,20}.npz`, `sqdpen_sq_it4.npz` (the projected-square point) and `sqdpen_true_it4.npz` (the literal-square point and its λ -sweep), each with final vectors; figure `fig_penalty_frontier.png` (`_sqdpenfig.py`).

Completion-mechanism audit (Section 2.6). `_degen.py` rebuilds every spin-completion checkpoint of the three trajectory arms from the archived determinant sets (`s2audit_*.npz`), re-solves the uncompleted and completed spaces with PyCI (lowest three roots, tolerance 10^{-9}), and records per checkpoint: the ground-energy shift under completion; the completed ground-manifold splitting; the invariant spectrum of the original-support projector compressed over the ground manifold; the localized pair member’s weight on the added determinants; the coupling $\|P_+ H \nu\|$ (the uncompleted ground embedded with zeros, the completed-space operator applied, the result projected onto the added determinants); and the invariant S^2 Gram block over the manifold (`degen.npz`). At the one checkpoint whose iterative record lacks a resolved pair ([4Fe-4S], 1,106 $\rightarrow$ 2,097), `_degen_dense.py` builds the full 2,097-dimensional Hamiltonian by explicit matrix–vector columns and diagonalizes it densely (LAPACK), restoring the exact degenerate pair and re-deriving the same row quantities gauge-invariantly; the iterative artifact (three distinct Ritz values, one per degenerate pair) is preserved in the same archive (`degen_dense.npz`). Table rows and prose bounds are generated from these archives by `_degen_table.py`.

Version matrix. `_verm.py` repeats the construction–solver–diagnostic semantics at the earliest publicly auditable release (0.3.0; we could not retrieve the stated 0.1.0 through PyPI or a tagged GitHub release as of 2026-07-15; Limitations item 7) and at 0.12.1, on identical inputs: the $N = 60$ reproducer core, and a seed-17, 500-shot in-sector batch of the raw [2Fe-2S] hardware counts with closure on and off, each version through its own en-

try points (bitstring_matrix_to_sorted_addresses / solve_fermion(..., max_davidson=400) at 0.3.0; bitstring_matrix_to_ci_strs / solve_fermion(..., max_cycle=400) at 0.12.1). Outputs verm_0_3_0.npz / verm_0_12_1.npz.

Flagship-dimension leg (Section 2.10). Outer-product subspaces from the top- N of the 7,658 released unique half-configurations per spin, ranked by the exact marginals of the rebuilt optimized-circuit state (lucjopt_marginals.npz; the released strings load as uint64 and must be cast to int64 and sorted before use). Solver: qiskit_addon_sqd.fermion.solve_fermion on their MO FCIDUMP, as shipped, cold-started, with max_cycle=400 and verbose=5 forwarded to PySCF (the shipped default max_cycle would return an under-converged flagship energy without warning; no other kwarg is set and spin_sq is left unset so the reported $\langle S^2 \rangle$ is their code’s own). Rungs $N \in \{300, 500, 600, 1000, 2000\}$ are certified against PyCI anchors on the identical construction (ΔE 0.002–0.023 mHa; the $n = 600$ rung also reproduces the exact cross-implementation reconstruction energy to 0.08 μ Ha). The $N = 7500$ rung (their published 5.625×10^7) ran 63,846 s at 32 threads on a 64-vCPU c7i.16xlarge (peak RSS 13.6 GB), 400 Davidson iterations to final residual 3.5×10^{-4} . Full amplitude matrices and string sets are archived in flagsolve_n{N}.npz; the $N = 2000$ rung was additionally converged on two further platform/thread configurations (flagsolve_n2000_box1.npz), yielding the $\langle S^2 \rangle$ run-instability datum; per-attempt cloud logs (flagsolve_box1_attempt{1,2,3}_live.log) document three spot-instance interruptions with no data loss (2-minute result syncing). The flagship rung was subsequently certified by a dedicated three-root block eigensolve of the identical subspace (pyscf.fci.selected_ci.kernel.fixed_space, nroots = 3; the same matrix-free kernel solve_fermion wraps), warm-started from the banked flagship vector under three pre-registered gates (bit-identical string sets; warm-vector Rayleigh quotient vs banked energy, measured 0.0 μ Ha; warm $\langle S^2 \rangle$ vs banked, $< 10^{-14}$), initial block = warm root plus two lowest-diagonal guesses, tolerance ladder $10^{-6}/10^{-7}/10^{-8}$ (max_cycle 150/150/200), with the S^2 -Gram audit and a full checkpoint after each stage. All three roots met the solver’s per-root convergence criteria, final residuals $(6.0, 8.4, 9.1) \times 10^{-5}$ (the tightest still 21% of the 0.285 mHa ground gap; see the stability-vs-residual discussion in Section 2.10); 11.3 h at 64 threads on a 64-core c7a.16xlarge, peak RSS 41.8 GB. Script _gram7500.py, log gram7500.run.log, per-stage summaries gram7500_ck{1,2,3}.npz, final gram7500.npz; the converged three-root vector block gram7500_restart.npz (1.3 GB) ships as a Zenodo companion beside the flagship amplitudes. The stage-4 follow-up (_gram7500s4.py; 14.8 h at 64 threads on the same instance class, peak RSS 43.7 GB) warm-starts from that certified block under the same gates plus an in-box moment gauge (H_4 /STO-3G FCI: per-eigenstate $|\text{Var}(S^2)| < 10^{-9}$ and an analytic two-state mixture variance reproduced to 10^{-9} , asserted before production; standalone version _s4gate.py). It grows the block with orthogonalized lowest-diagonal guesses and re-solves through stages (nroots, tol, max_cycle) = (3, 10^{-10} , 120), (4, 10^{-8} , 150), (4, 10^{-10} , 150), re-measuring energies, residuals, the S^2 -Gram, and the S^4 moments after every stage (checkpoints gram7500_s4ck{0-3}.npz, final gram7500_s4.npz, run log gram7500s4_live.log; the final four-root vector block gram7500_s4_restart.npz, 1.8 GB, ships as a Zenodo companion beside the three-root block). Second moments use the $M_s = 0$ identity: with $W = S^+V$ the deduplicated S^+ -image of the root block and $W_2 = S^+W$, $G = W^T W$ and $G_4 = W_2^T W_2 + 2G$, so $\text{Var}(S^2)$ per root is $(G_4)_{ii} - (G)_{ii}^2$; the $\{S = 0, 1, 2\}$ sector solve inverts the moment pair on the sector eigenvalues $\{0, 2, 6\}$ ($w_2 = (\langle S^4 \rangle - 2\langle S^2 \rangle^2)/24$, $w_1 = (6\langle S^2 \rangle - \langle S^4 \rangle)/8$, $w_0 = 1 - w_1 - w_2$), a negative weight certifying support outside $S \leq 2$. The same gauged moment kernel is applied by _s4states.py to the eight final pipeline states of the cross-instrument legs (Section 2.11), each gated against the shipped spin_square to 10^{-6} before its $\langle S^4 \rangle$ and sector solve are recorded (s4states.npz). A follow-on kernel (_s6moments.py) extends the same $M_s = 0$ S^+ -ladder on those eight states to the third and fourth moments, $\langle S^6 \rangle = |\phi_3|^2 + 8|\phi_2|^2 + 4|\phi_1|^2$ and $\langle S^8 \rangle = |\phi_4|^2 + 20|\phi_3|^2 + 52|\phi_2|^2 + 8|\phi_1|^2$, with $\phi_k = (S^+)^k \psi$ built by chunked deduplication. It was checked in-process in two ways before production, both hard-asserted. On H_4 /STO-3G FCI, every eigenstate closes as $\langle S^{2k} \rangle = \lambda^k$ and an analytic 0.7/0.3 singlet–triplet mixture reproduces $(\langle S^6 \rangle, \langle S^8 \rangle) = (2.4, 4.8)$ to $< 10^{-9}$; and on pure maximum-spin $S = 3, 4, 5$ $M_s = 0$ reference states (built as the top eigenvector of

PySCF’s own `contract_ss` S^2 operator, independent of the S^+ kernel), every measured moment closes as $\langle S^{2k} \rangle = [S(S+1)]^k$ ($k = 1 \dots 4$) to $< 10^{-6}$ relative. This second gauge exercises the third and fourth raising steps ϕ_3/ϕ_4 , which the H4 gauge (whose FCI space caps at $S = 2$) never drives with nonzero vectors, so the [4Fe-4S] LP (which runs to $S = 9$) rests on verified moments at every rung it uses. The measured moments then constrain a linear program (`scipy.optimize.linprog`, HiGHS) over sector weights $w_S \geq 0$ on the full physical grid $S = 0 \dots S_{\max}$ ($S_{\max} = (2 \text{ ncas} - \text{nelec})/2$: 5 for [2Fe-2S], 9 for [4Fe-4S]), with the norm and each available moment enforced to tolerance $10^{-7} \cdot \text{scale}$. The LP returns numerically certified minimum and maximum values of the singlet, triplet, and combined $S \geq 3$ weights (throughout, “numerically certified” denotes the min/max over the LP-feasible set at these tolerances, not an analytic bound), plus the unique $\{0, 1, 2, 3\}$ four-moment solve whose negative component would certify $S \geq 4$ support (`s6moments.npz`). Each state’s m_1 is re-gated against the shipped `spin_square` (10^{-6}) and its (m_1, m_2) against the frozen `s4states.npz` values ($10^{-8}/10^{-6}$) before its bounds are recorded. A sensitivity companion (`_s6stress.py`) re-runs the identical LP with every per-moment box inflated 10 $\times$, 100 $\times$, and 1000 $\times$, gating the factor-1 run against the production bounds before recording the inflated ones (`s6stress.npz`). Roots 0–2 meet the solver’s per-root criteria at every stage (final measured residuals $(3.2, 4.1, 2.6) \times 10^{-6}$); root 3 reaches the final stage’s iteration cap with measured residual 3.3×10^{-5} (energy stable to $< 10^{-9}$ Ha per iteration over the closing ten); we therefore report its energy and moments as measured properties of the returned vector, without labeling that root converged.

Independent-implementation flagship cross-check (Section 2.10). `_tb7500box.py` on a storage-optimized cloud instance (`i4i.16xlarge`: 64 vCPU, 512 GiB RAM, 4×3.75 TB NVMe in RAID-0). The $N = 7500$ outer-product subspace is rebuilt from `tb7500_strings.npz` (hash-gated against the flagship payload), and the Hamiltonian over it is constructed by PyCI in 512 row-chunks across 20 worker processes (rectangular `sparse_op` rows under a closed-form column map; global lower triangle, each pair stored exactly once), streamed to a 1.6 TB on-disk CSR triangle. Pre-registered gates, in order: payload hashes; chunk-0 rebuilt bit-identically by parent and worker; assembled nonzero count equal to the exact combinatorial prediction, 127,975,471,122, an identity rather than an estimate, reproduced exactly by a second full build on a second fresh instance; Rayleigh quotients of the certified stage-4 three-root block in the independent operator vs the certified energies (measured 1.4×10^{-14} , 2.0×10^{-13} , 1.3×10^{-13} Ha); post-solve energy and $\langle S^2 \rangle$ agreement (tolerances 5×10^{-6} Ha, 5×10^{-3}). The eigensolve is an in-house block Davidson (three roots, diagonal preconditioner, subspace cap 48, per-iteration checkpointing to the instance NVMe with 2-minute off-box sync) over threaded symmetric-triangle matrix–vector kernels (64 threads, ~ 570 s per block iteration, operator streamed from NVMe), warm-started from the certified block under a deliberate 10^{-5} -scale per-component jitter; $\langle S^2 \rangle$ is measured by embedding each converged root into the full $M_s = 0$ sector (1.92 GB per root) through PySCF’s `spin_square`. 38 iterations to maximum residual 1.25×10^{-5} at the pre-set wall budget, ≈ 9.5 h wall. Artifacts: `tb7500_result_n7500.npz` (energies, residuals, $\langle S^2 \rangle$, warm Rayleigh quotients, nonzero count), the three-root vector block `tb7500_ck.npz` (1.35 GB; ships as a Zenodo companion beside the kernel-side blocks), and the run logs of both builds. A follow-on warm-resume attempt was terminated after a diagnosed checkpoint-format limitation (the per-iteration checkpoint stores the Ritz block but not the accumulated Davidson subspace, and a subspace-free restart stalls against the 0.285 mHa pair), retaining its exact-count second build and Rayleigh-quotient gates; the stall and termination are documented in the retained logs.

Archive audit (Sections 2.7–2.11). All IBM-archive inputs are consumed as released (zenodo.15324153). Their [2Fe-2S] hardware file stores 300 per-batch dictionaries of probabilities (each summing to 1.0 over one 8,192-shot batch); we convert to integer shot counts as `round( $v \times 8192$ )` with a per-batch total assertion ($300 \times 8192 = 2,457,600$, matching the published shot budget). Their [4Fe-4S] file is a single normalized outcome dictionary in which every value equals $1/3,163,742$ (every recorded outcome appeared exactly once) converted to unit counts. Their noiseless optimal-circuit samples are boolean occupation matrices whose columns run orbital $19 \rightarrow 0$ within each spin half; packing is gated on the archive’s own documented invariant

that row 0 is the Hartree–Fock string. Uniform-control, HCI-ladder, headline-energy, and recovery-ablation text files are parsed verbatim (`_ibmuniform.py`, audit section J). Energies are referenced to their tabulated DMRG values.

References (energies). [2Fe-2S] exact singlet -116.6056091 Ha (near-exact reference distributed with the FCIDUMP set, and identical to the DMRG line in IBM’s own archive); [4Fe-4S] -327.2396369 Ha (DMRG $d = 4000$, approximate per the source repository, used only as a distance scale, never as an exactness claim).

Automated claim verification. Every quantitative statement in this paper (table rows, prose values, ratios, sector arithmetic) is re-derived from the raw result archives by an audit script (`_paperaudit.py`, supplementary archive) that fails loudly on any mismatch; the version of record passes with zero failures. This audit corrected several memory-carried values in an earlier draft: the same discipline, applied to the manuscript itself. Its epistemic scope should be stated plainly: a passing audit certifies internal consistency (agreement between the manuscript’s numbers and its own archived artifacts, under expectations we encoded), not the scientific correctness of those expectations, and it is not a substitute for independent external reproduction, which the archive exists to enable.

5 Limitations

We list the limitations a critical reader would raise, including four from earlier drafts that are now closed. The first two concerned provenance: that the integrals and the subspaces were ours rather than IBM’s. Sections 2.7–2.11 now audit IBM’s own MO-basis FCIDUMPs, whose spectrum we showed to be identical to that of the benchmark dumps, and IBM’s raw hardware shots, the ideal samples of its circuit template and its published configuration set. The third was that the flagship point might be under-converged. The three-root certification run of Section 2.10, completed before release, moved the delivered $\langle S^2 \rangle$ of 1.287 to a certified 1.371, which measured the conditioning that the gap implies, and a stage-4 ladder has since taken the residual to 3.2×10^{-6} , 1.1% of the ground gap, with the value unchanged at 1.3711. The fourth was that the certification relied on a single kernel checking itself. The independent-implementation run of Section 2.10, a PyCI-built terabyte-scale operator with an in-house block eigensolver and no PySCF selected-CI component, reproduces the exact combinatorial nonzero count twice on fresh instances, the certified energies to $\leq 8 \times 10^{-10}$ Ha and the certified $\langle S^2 \rangle$ ladder to $\leq 7.5 \times 10^{-4}$, at a residual of 1.25×10^{-5} (4.4% of the ground gap) whose rotation bounds contain every observed deviation. The following limitations remain.

1. **Scale of the spin-audited subspaces.** For [2Fe-2S], the audit reaches the largest published dimension. The certified flagship-dimension point of Section 2.10 measures $\langle S^2 \rangle$ at 5.625×10^7 determinants with a three-root solve of the subspace reconstructed here at the published construction. It is numerically stable across two independent tolerance ladders (final measured ground residual 3.2×10^{-6} , 1.1% of the 0.285 mHa ground gap; lowest root $\langle S^2 \rangle = 1.3711$, unchanged in the fourth decimal, with second moments certifying spin support beyond $S = 2$), and it is confirmed by a fully independent implementation (energies to $\leq 8 \times 10^{-10}$ Ha and the $\langle S^2 \rangle$ ladder to $\leq 7.5 \times 10^{-4}$, at a residual of 1.25×10^{-5} stated with its rotation bounds in Section 2.10). Two gaps remain. (i) That point uses the idealized exact-marginal construction. The shot-level ten-batch recovery protocol itself is audited only to 4×10^6 , a factor of 14 below the top, so recovery dynamics at 5.6×10^7 are inferred, not measured. The inference runs in one direction: the idealized limit is the pipeline’s best case, and the completed raw-data ladder moves away from the singlet as dimension grows ($\langle S^2 \rangle$ 0.11 $\rightarrow$ 0.36). (ii) The [4Fe-4S] runs at 10^8 and the successor’s 10^9 – 10^{10} (arXiv:2511.00224, with a larger [2Fe-2S] active space) remain beyond audit, although the uniform-control energetics released with them still bracket those dimensions (Section 2.7).

2. **[4Fe-4S] reference.** The -327.2396 Ha reference is approximate (the archive labels it DMRG). We therefore claim only the $\langle S^2 \rangle$ trend, which needs no reference, the multi-spin manifold structure, and relative comparisons made entirely within the archive’s own files (SQD against HCI against uniform).
3. **The optimal-circuit sampling distribution.** Because the published 58.6M-configuration file is a deduplicated set (Section 2.9), the shot-level distribution of the optimal-circuit numerics cannot be audited from public data. Our statements about that leg concern the set, the rebuilt state and the pipeline, not the unreleased shot record.
4. **Template-leg CCSD amplitudes.** The CCSD used to parameterize our noiseless template leg carried a non-convergence flag ($E_{\text{corr}} = -0.20026$ against the tabulated -0.19985 , a 0.4 mHa difference). This affects only the ansatz initialization of one leg, and none of that leg’s conclusions depend on amplitude precision.
5. **Exact verification.** The matched-dimension conclusion of Section 2.1 is verified against exact results only where FCI is reachable. On the flagship systems it rests on matched-dimension comparisons against the sampling parent (Section 2.6) and on relative comparisons within IBM’s own archive (Sections 2.7–2.11), not on exact answers.
6. **Penalty provenance at production scale.** The flagship Methods state that the soft constraint was employed at $\lambda = 0.2$, but the released archive does not record per-run eigensolver settings, so whether and where the production solves enabled it cannot be verified from public artifacts. Our audit therefore reproduces the stated form and strength on the subspaces that public data allow us to reconstruct (Section 2.6), and on those the constraint does not produce the named singlet at any competitive energy. Separately, the released package’s corresponding `spin_sq` argument defaults to unset (item 7).
7. **Package version.** The as-shipped runs in this paper use `qiskit-addon-sqd 0.12.1`, the release current at the time of the audit; the flagship paper states 0.1.0. That stated version is not publicly retrievable: PyPI’s earliest release is 0.3.0 and the GitHub repository has no 0.1.0 tag (both checked 2026-07-15), although the archive README says all versions are available on both, so the published semantics can be audited only from 0.3.0 onward. We therefore ran a version matrix at the earliest publicly auditable release (Methods; `_verm.py`). On identical inputs (the $N = 60$ reproducer core, and a seed-matched 500-shot in-sector batch of the raw hardware counts with spin-inversion closure on and off), the construction, solver and spin diagnostic of 0.3.0 reproduce the 0.12.1 energies and $\langle S^2 \rangle$ to all printed digits (sub- μ Ha; reproducer-core $\langle S^2 \rangle = 3.5009$ through both versions’ own diagnostics) at identical subspace dimensions. The only difference observed is a swapped α/β half-ordering convention in the closure-off construction, which is physically equivalent. The spin-penalty argument `spin_sq` exists with default `None`, connected to PySCF’s `fix_spin_`, in both versions, so the unset-by-default penalty measured here is a property of the public release history, not of one release. The 35-second reproducer is also independent of the package (pure PySCF on the public data).

AI-use disclosure

An AI research agent (Claude, Anthropic) carried out the computations and tooling reported here, under the author’s direction; the research program, the methodological and scope decisions, and the final text are the author’s. Agent-executed work should not be taken on trust. The manuscript is therefore built to be checked: `_paperaudit.py` re-derives every quantitative claim from the raw result archives (370 checks, zero failures), and the agent’s full self-audit record, including five self-caught defects, one self-retraction, two reviewer-prompted corrections, and one post-publication correction, is preserved in the supplementary audit trail (`AUDIT_TRAIL.md`).

Data and code availability

A DOI-archived supplementary package accompanies this paper (Zenodo: [10.5281/zenodo.21359923](https://doi.org/10.5281/zenodo.21359923); the flagship-dimension amplitude archive `flagsolve_n7500_box2.npz` and the certification vector blocks `gram7500_restart.npz` (three-root), `gram7500_s4_restart.npz` (four-root, stage 4), and `tb7500_ck.npz` (three-root, independent implementation) ship there as standalone companion files next to the zip): `s2audit_master.csv` (full audit table), figures (`fig_s2_vs_d.png`, `fig_err_vs_d.png`, `fig_coupon.png`, `fig_flagship_s4.png`, `fig_penalty_frontier.png`), per-run archives `s2audit_*.npz`, as-shipped trajectories (`sqdship_*.npz`, `ibmship_*.npz`, `lucj2ship_*.npz`, `lucjoptship_*.npz`, `ibm4ship_*.npz`) with their counts and statistics, the audit engine `_s2audit.py`, the harnesses (`_sqdship.py`, `_ibmsamples.py`, `_ibmsamples4.py`, `_lucjfe2.py`, `_lucjopt.py`), the sampling-law and cost-model scripts (`_couponlaw.py`, `_costmodel.py`, `_ibmuniform.py`, and the batch-statistics script `_ibmuniform2.py` with its archive `ibmuniform_stats.npz`), the standalone ~35-second reproducer `spin_inversion_is_not_singlet.py` (Section 2.6), the flagship multi-root certification (`_gram7500.py`, `gram7500.npz`, `gram7500_ck{1,2,3}.npz`, `gram7500.run.log`), the stage-4 second-moment certification (`_gram7500s4.py`, `_s4gate.py`, `gram7500_s4ck{0-3}.npz`, `gram7500_s4.npz`, `gram7500s4_live.log`), the independent-implementation flagship cross-check (`_tb7500box.py`, `tb7500_strings.npz`, `tb7500_result_n7500.npz`, both builds’ run logs), the eight-state cross-instrument second-moment audit (`_s4states.py`, `s4states.npz`), the higher-moment and sector-bound audit with its sensitivity companion (`_s6moments.py`, `s6moments.npz`, `_s6stress.py`, `s6stress.npz`), the penalty-on audit (`_sqdpen.py`, `_sqdpen_sq.py`, `_sqdpen_true.py`, `_sqdpen_diagprobe.py`, `_sqdpen_s3b.py`, `_sqdpenfig.py`, `sqdpen_frontier.npz`, `sqdpen_theirs.npz`, `sqdpen_spaces.npz`, `sqdpen_loop_l{10,20}.npz`, `sqdpen_s3b.npz`, `sqdpen_sq_it4.npz`, `sqdpen_true_it4.npz`, `sqdpen_true_it1.npz`, `_sqdpen.run.log`, `_sqdpen_sq.run.log`, `_sqdpen_true.run.log`), the completion-mechanism audit (`_degen.py`, `_degen_dense.py`, `_degen_flip.py`, `_degen_table.py`, `degen.npz`, `degen_dense.npz`, `_degen.run.log`), the claim-verification script `_paperaudit.py`, and a claim-by-claim reproduction map (REPRO_MAP.md) giving, for every headline claim, the artifact that proves it and the command that regenerates it. Inputs: the public Li–Chan active-space FCIDUMPs; IBM’s own published data archive for arXiv:2405.05068 (DOI 10.5281/zenodo.15324153), from which we consume the raw hardware counts, MO-basis integrals, circuit template and parameters, noiseless optimal-circuit samples, uniform controls, HCI references, and recovery-ablation files exactly as released; and the publicly released `qiskit-addon-sqd` (run as shipped) and `PyCI` packages. Full AI session logs supporting the audit trail are available from the author on request.

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Prior work is cited inline in the text by arXiv identifier or DOI; the consolidated list follows. arXiv identifiers resolve to the canonical record for each preprint.

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8. Software run as shipped: IBM `qiskit-addon-sqd` (releases 0.3.0–0.12.1), PyCI, and PySCF (FCI / selected-CI kernels and spin operators).
9. This work. DOI-archived supplementary package. DOI [10.5281/zenodo.21359923](https://doi.org/10.5281/zenodo.21359923).